

LAPLACE INVERSE

Type I: Basic

1. Find Inverse Laplace Transforms of $\frac{2}{s} + \frac{1}{s^3} + \frac{1}{s+4}$

Ans. $2 + \frac{t^2}{2} + e^{-4t}$

2. Find the inverse L.T. of $\frac{3+2s+s^2}{s^3}$

[M19/Chem/3M]

Solution:

$$\begin{aligned} L^{-1} \left\{ \frac{3+2s+s^2}{s^3} \right\} &= L^{-1} \left\{ \frac{3}{s^3} + \frac{2s}{s^3} + \frac{s^2}{s^3} \right\} \\ &= L^{-1} \left\{ \frac{3}{s^3} + \frac{2}{s^2} + \frac{1}{s} \right\} \\ &= 3 \cdot \frac{t^2}{2!} + 2 \cdot t + 1 \end{aligned}$$

3. Find Inverse Laplace Transforms of $\frac{3s+4}{s^2+16}$

Ans. $3\cos 4t + \sin 4t$

4. Find the inverse Laplace of $\left\{ \frac{s+a}{s^2+16} \right\}$ and find the constant a if $f\left(\frac{\pi}{8}\right) = 1$

[N19/Elect/4M]

Solution:

$$L^{-1} \left\{ \frac{s+a}{s^2+16} \right\} = L^{-1} \left\{ \frac{s}{s^2+16} + \frac{a}{s^2+16} \right\} = \cos 4t + a \frac{\sin 4t}{4}$$

Now,

$$f(t) = \cos 4t + a \frac{\sin 4t}{4}$$

Put $t = \frac{\pi}{8}$

$$f\left(\frac{\pi}{8}\right) = \cos \frac{\pi}{2} + \frac{a \sin \frac{\pi}{2}}{4}$$

$$1 = 0 + \frac{a}{4}$$

$$\therefore a = 4$$

5. Find Inverse L.T. of: $\left(\frac{1-\sqrt{s}}{s^2}\right)^2$

Ans. $\frac{t^3}{6} - \frac{16}{15\sqrt{\pi}} \cdot t^{\frac{5}{2}} + \frac{t^2}{2}$

6. Find Inverse Laplace Transforms of: $\left[\frac{3s-2}{s^{\frac{5}{2}}} - \frac{7}{3s+2} \right]$

Ans. $\frac{6t^{\frac{1}{2}}}{\sqrt{\pi}} - 8 \cdot \frac{t^{\frac{3}{2}}}{3\sqrt{\pi}} - \frac{7}{3} e^{-\frac{2}{3}t}$

7. Find Inverse Laplace Transforms of: $\left[\frac{3s-8}{s^2+4} - \frac{4s-24}{s^2-16} \right]$

Ans. $3\cos 2t - 4\sin 2t - 4\cosh 4t + 6\sinh 4t$

8. Find Inverse Laplace Transforms of: $\frac{s+1}{s^2-4}$

Ans. $\cosh 2t + \frac{\sinh 2t}{2}$



Type II: First Shifting Property

1. Find Inverse Laplace Transforms of: $\frac{s+2}{s^2+4s+7}$ Ans. $e^{-2t} \cdot \cos \sqrt{3}t$

2. Find Inverse Laplace Transforms of: $\frac{s+2}{s^2-2s+17}$

[M16/ElexExtcElectBiomInst/4M]

Solution:

$$\begin{aligned} L^{-1} \left\{ \frac{s+2}{s^2-2s+17} \right\} &= L^{-1} \left\{ \frac{s+2}{s^2-2s+1-1+17} \right\} \\ &= L^{-1} \left\{ \frac{s+2}{(s-1)^2+16} \right\} \\ &= L^{-1} \left\{ \frac{s-1+1+2}{(s-1)^2+16} \right\} \\ &= e^t L^{-1} \left\{ \frac{s+3}{s^2+16} \right\} \\ &= e^t L^{-1} \left\{ \frac{s}{s^2+16} + \frac{3}{s^2+16} \right\} \\ &= e^t \left[\cos 4t + \frac{3}{4} \sin 4t \right] \end{aligned}$$

3. Find inverse LT of $\frac{s-1}{s^2+2s+2}$

[N19/Comp/4M]

Solution:

$$\begin{aligned} L^{-1} \left\{ \frac{s-1}{s^2+2s+2} \right\} &= L^{-1} \left\{ \frac{s-1}{s^2+2s+1-1+2} \right\} \\ &= L^{-1} \left\{ \frac{s-1}{(s+1)^2+1} \right\} \\ &= L^{-1} \left\{ \frac{s+1-1-1}{(s+1)^2+1} \right\} \\ &= e^{-t} L^{-1} \left\{ \frac{s-2}{s^2+1} \right\} \\ &= e^{-t} L^{-1} \left\{ \frac{s}{s^2+1} - \frac{2}{s^2+1} \right\} \\ &= e^{-t} [\cos t - 2 \sin t] \end{aligned}$$

4. Find Inverse Laplace Transforms of: $\frac{s}{(2s+1)^2}$

[N13/CompIT/3M]

Solution:

$$\begin{aligned} L^{-1} \left\{ \frac{s}{(2s+1)^2} \right\} &= \frac{1}{4} L^{-1} \left\{ \frac{s}{\left(s+\frac{1}{2}\right)^2} \right\} \\ &= \frac{1}{4} L^{-1} \left\{ \frac{s+\frac{1}{2}-\frac{1}{2}}{\left(s+\frac{1}{2}\right)^2} \right\} \\ &= \frac{1}{4} e^{-\left(\frac{1}{2}t\right)} L^{-1} \left\{ \frac{s-\frac{1}{2}}{s^2} \right\} \end{aligned}$$

$$\begin{aligned}
 &= \frac{e^{-\left(\frac{t}{2}\right)}}{4} L^{-1} \left\{ \frac{s}{s^2} - \frac{\frac{1}{2}}{s^2} \right\} \\
 &= \frac{e^{-\frac{t}{2}}}{4} L^{-1} \left\{ \frac{1}{s} - \frac{1}{2s^2} \right\} \\
 &= \frac{e^{-\frac{t}{2}}}{4} \left[1 - \frac{t}{2} \right]
 \end{aligned}$$

5. Find Inverse Laplace Transform of: $\frac{s}{(s-2)^6}$

[N14/AutoMechCivil/4M]

Solution:

$$\begin{aligned}
 L^{-1} \left\{ \frac{s}{(s-2)^6} \right\} &= L^{-1} \left\{ \frac{s-2+2}{(s-2)^6} \right\} \\
 &= e^{2t} L^{-1} \left\{ \frac{s+2}{s^6} \right\} \\
 &= e^{2t} L^{-1} \left\{ \frac{s}{s^6} + \frac{2}{s^6} \right\} \\
 &= e^{2t} L^{-1} \left\{ \frac{1}{s^5} + \frac{2}{s^6} \right\} \\
 &= e^{2t} \left[\frac{t^4}{4!} + \frac{2t^5}{5!} \right] \\
 &= e^{2t} \left[\frac{t^4}{24} + \frac{t^5}{60} \right]
 \end{aligned}$$

6. Find Inverse Laplace Transforms of: $\frac{s^2}{(s+a)^3}$ Ans. $e^{-at} \left[1 - 2at + \frac{a^2 t^2}{2} \right]$

7. Find Inverse Laplace Transforms of: $\frac{s+2}{s^2-4s+13}$

[M15/ElexExtcElectBiomInst/4M][N16/CompIT/4M]

Solution:

$$\begin{aligned}
 L^{-1} \left\{ \frac{s+2}{s^2-4s+13} \right\} &= L^{-1} \left\{ \frac{s+2}{s^2-4s+4-4+13} \right\} \\
 &= L^{-1} \left\{ \frac{s+2}{(s-2)^2+9} \right\} \\
 &= L^{-1} \left\{ \frac{s-2+2+2}{(s-2)^2+9} \right\} \\
 &= e^{2t} L^{-1} \left\{ \frac{s+4}{s^2+9} \right\} \\
 &= e^{2t} L^{-1} \left\{ \frac{s}{s^2+9} + \frac{4}{s^2+9} \right\} \\
 &= e^{2t} \left[\cos 3t + 4 \cdot \frac{\sin 3t}{3} \right] \\
 &= e^{2t} \cos 3t + \frac{4}{3} e^{2t} \sin 3t
 \end{aligned}$$

8. Find $L^{-1} \left\{ \frac{5s+3}{s^2+6s+25} \right\}$

[N17/Elect/Biom/4M][M19/Inst/4M]



Solution:

$$\begin{aligned}
 L^{-1} \left\{ \frac{5s+3}{s^2+6s+25} \right\} &= 5L^{-1} \left\{ \frac{s+\frac{3}{5}}{s^2+6s+9-9+25} \right\} \\
 &= 5L^{-1} \left\{ \frac{s+3-3+\frac{3}{5}}{(s+3)^2+16} \right\} \\
 &= 5e^{-3t} L^{-1} \left\{ \frac{s-\frac{12}{5}}{s^2+16} \right\} \\
 &= 5e^{-3t} L^{-1} \left\{ \frac{s}{s^2+16} - \frac{\frac{12}{5}}{s^2+16} \right\} \\
 &= 5e^{-3t} \left[\cos 4t - \frac{12}{5} \cdot \frac{\sin 4t}{4} \right] \\
 &= e^{-3t} [5 \cos 4t - 3 \sin 4t]
 \end{aligned}$$

9. Find $L^{-1} \left[\frac{2s-1}{s^2+4s+29} \right]$

[N15/ElexExtcElectBiomInst/4M]

Solution:

$$\begin{aligned}
 L^{-1} \left\{ \frac{2s-1}{s^2+4s+29} \right\} &= 2L^{-1} \left\{ \frac{s-\frac{1}{2}}{s^2+4s+4-4+29} \right\} \\
 &= 2L^{-1} \left\{ \frac{s-\frac{1}{2}}{(s+2)^2+25} \right\} \\
 &= 2L^{-1} \left\{ \frac{s+2-2-\frac{1}{2}}{(s+2)^2+25} \right\} \\
 &= 2e^{-2t} L^{-1} \left\{ \frac{s-\frac{5}{2}}{s^2+25} \right\} \\
 &= 2e^{-2t} L^{-1} \left\{ \frac{s}{s^2+25} - \frac{\frac{5}{2}}{s^2+25} \right\} \\
 &= 2e^{-2t} \left[\cos 5t - \frac{5}{2} \cdot \frac{\sin 5t}{5} \right] \\
 &= 2e^{-2t} \left[\cos 5t - \frac{\sin 5t}{2} \right]
 \end{aligned}$$

10. Find $L^{-1} \left\{ \frac{s+6}{(s-5)^2+121} \right\}$

[N19/Elect/4M]

Solution:

$$\begin{aligned}
 L^{-1} \left\{ \frac{s+6}{(s-5)^2+121} \right\} &= L^{-1} \left\{ \frac{s-5+5+6}{(s-5)^2+121} \right\} \\
 &= e^{5t} L^{-1} \left\{ \frac{s+11}{s^2+121} \right\} \\
 &= e^{5t} L^{-1} \left\{ \frac{s}{s^2+121} + \frac{11}{s^2+121} \right\} \\
 &= e^{5t} [\cos 11t + \sin 11t]
 \end{aligned}$$



11. Find $L^{-1} \left[\frac{s+5}{(s+4)^3} \right]$

[N19/Elex/3M]

Solution:

$$\begin{aligned} L^{-1} \left[\frac{s+5}{(s+4)^3} \right] &= L^{-1} \left[\frac{s+4+1}{(s+4)^3} \right] \\ &= e^{-4t} L^{-1} \left\{ \frac{s+1}{s^3} \right\} \\ &= e^{-4t} L^{-1} \left\{ \frac{s}{s^3} + \frac{1}{s^3} \right\} \\ &= e^{-4t} L^{-1} \left\{ \frac{1}{s^2} + \frac{1}{s^3} \right\} \\ &= e^{-4t} \left[t + \frac{t^2}{2} \right] \end{aligned}$$

12. Find $L^{-1} \left[\frac{6}{(2s+1)^3} \right]$

[M17/CompIT/4M]

Solution:

$$\begin{aligned} L^{-1} \left\{ \frac{6}{(2s+1)^3} \right\} &= \frac{6}{8} L^{-1} \left\{ \frac{1}{\left(s+\frac{1}{2}\right)^3} \right\} \\ &= \frac{3}{4} e^{-\frac{1}{2}t} L^{-1} \left\{ \frac{1}{s^3} \right\} \\ &= \frac{3}{4} e^{-\left(\frac{1}{2}t\right)} \cdot \frac{t^2}{2!} = \frac{3}{8} e^{-\frac{t}{2}} t^2 \end{aligned}$$

13. Find the inverse L.T. of $\frac{3s+1}{(s+1)^4}$

[N18/Comp/4M]

Solution:

$$\begin{aligned} L^{-1} \left\{ \frac{3s+1}{(s+1)^4} \right\} &= 3L^{-1} \left\{ \frac{s+\frac{1}{3}}{(s+1)^4} \right\} \\ &= 3L^{-1} \left\{ \frac{s+1-1+\frac{1}{3}}{(s+1)^4} \right\} \\ &= 3e^{-t} L^{-1} \left\{ \frac{s-\frac{2}{3}}{s^4} \right\} \\ &= 3e^{-t} L^{-1} \left\{ \frac{s}{s^4} - \frac{\frac{2}{3}}{s^4} \right\} \\ &= 3e^{-t} L^{-1} \left\{ \frac{1}{s^3} - \frac{2}{3s^4} \right\} \\ &= 3e^{-t} \left[\frac{t^2}{2!} - \frac{2}{3} \cdot \frac{t^3}{3!} \right] \\ &= 3e^{-t} \left[\frac{t^2}{2} - \frac{t^3}{9} \right] \end{aligned}$$

14. Find $L^{-1} \left[\frac{1}{\sqrt{2s+1}} \right]$

[N17/Comp/5M][N19/IT/4M]

Solution:

$$\begin{aligned} L^{-1} \left\{ \frac{1}{\sqrt{2s+1}} \right\} &= \frac{1}{\sqrt{2}} L^{-1} \left\{ \frac{1}{\sqrt{s+\frac{1}{2}}} \right\} \\ &= \frac{1}{\sqrt{2}} e^{-\frac{1}{2}t} L^{-1} \left\{ \frac{1}{\sqrt{s}} \right\} \\ &= \frac{e^{-\frac{1}{2}t}}{\sqrt{2}} \cdot t^{-\frac{1}{2}} \\ &= \frac{e^{-\frac{1}{2}t}}{\sqrt{2} \sqrt{\pi} \sqrt{t}} \\ &= \frac{e^{-\frac{1}{2}t}}{\sqrt{2\pi t}} \end{aligned}$$

Type III: Method of Partial Fractions

1. Find Inverse Laplace Transforms of: $\frac{4s+12}{s^2+8s+12}$ Ans. $e^{-2t} + 3e^{-6t}$

2. Find Inverse Laplace Transforms of: $\frac{s^2+1}{s^3+3s^2+2s}$ Ans. $\frac{1}{2} - 2e^{-t} + \frac{5}{2}e^{-2t}$

3. Find Inverse Laplace Transforms of: $\frac{3s+7}{s^2-2s-3}$

[N17/Chem/6M][M18/Comp/5M]

Solution:

$$\text{Let } \frac{3s+7}{(s-3)(s+1)} = \frac{A}{s-3} + \frac{B}{s+1}$$

$$3s + 7 = A(s + 1) + B(s - 3)$$

Comparing the coefficients, we get

$$A + B = 3$$

$$A - 3B = 7$$

On solving the equations, we get

$$A = 4, B = -1$$

$$L^{-1} \left\{ \frac{3s+7}{s^2-2s-3} \right\} = L^{-1} \left\{ \frac{4}{s-3} - \frac{1}{s+1} \right\} = 4e^{3t} - e^{-t}$$

4. Find Inverse Laplace Transforms of: $\frac{8s+20}{s^2-12s+32}$

[N18/Elex/4M]

Solution:

$$\text{Let } \frac{8s+20}{(s-4)(s-8)} = \frac{A}{s-4} + \frac{B}{s-8}$$

$$8s + 20 = A(s - 8) + B(s - 4)$$



Comparing the coefficients, we get

$$A + B = 8$$

$$-8A - 4B = 20$$

On solving the equations, we get

$$A = -13, B = 21$$

$$L^{-1} \left\{ \frac{8s+20}{s^2-12s+32} \right\} = L^{-1} \left\{ \frac{-13}{s-4} + \frac{21}{s-8} \right\} = -13e^{4t} + 21e^{8t}$$

5. Find the inverse LT of $\frac{s}{(s-1)(s-2)(s-3)}$

[M19/Chem/3M]

Solution:

$$\text{Let } \frac{s}{(s-1)(s-2)(s-3)} = \frac{A}{s-1} + \frac{B}{s-2} + \frac{C}{s-3}$$

$$s = A(s-2)(s-3) + B(s-1)(s-3) + C(s-1)(s-2)$$

$$s = A(s^2 - 5s + 6) + B(s^2 - 4s + 3) + C(s^2 - 3s + 2)$$

Comparing the coefficients, we get

$$A + B + C = 0$$

$$-5A - 4B - 3C = 1$$

$$6A + 3B + 2C = 0$$

On solving the equations, we get

$$A = \frac{1}{2}, B = -2, C = \frac{3}{2}$$

$$L^{-1} \left\{ \frac{s}{(s-1)(s-2)(s-3)} \right\} = L^{-1} \left\{ \frac{\frac{1}{2}}{s-1} - \frac{2}{s-2} + \frac{\frac{3}{2}}{s-3} \right\} = \frac{e^t}{2} - 2e^{2t} + \frac{3e^{3t}}{2}$$

6. Find inverse LT of $\frac{2s^2-6s+5}{s^3-6s^2+11s-6}$

[N19/IT/4M]

Solution:

$$\text{Let } \frac{2s^2-6s+5}{(s-1)(s-2)(s-3)} = \frac{A}{s-1} + \frac{B}{s-2} + \frac{C}{s-3}$$

$$2s^2 - 6s + 5 = A(s-2)(s-3) + B(s-1)(s-3) + C(s-1)(s-2)$$

$$2s^2 - 6s + 5 = A(s^2 - 5s + 6) + B(s^2 - 4s + 3) + C(s^2 - 3s + 2)$$

Comparing the coefficients, we get

$$A + B + C = 2$$

$$-5A - 4B - 3C = -6$$

$$6A + 3B + 2C = 5$$

On solving the equations, we get

$$A = \frac{1}{2}, B = -1, C = \frac{5}{2}$$

$$L^{-1} \left\{ \frac{2s^2-6s+5}{s^3-6s^2+11s-6} \right\} = L^{-1} \left\{ \frac{\frac{1}{2}}{s-1} - \frac{1}{s-2} + \frac{\frac{5}{2}}{s-3} \right\} = \frac{e^t}{2} - e^{2t} + \frac{5e^{3t}}{2}$$



7. Find Inverse Laplace Transforms of: $\frac{s+2}{s^2(s+3)}$

[M19/Comp/5M]

Solution:

$$\text{Let } \frac{s+2}{s^2(s+3)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+3}$$

$$s + 2 = As(s + 3) + B(s + 3) + Cs^2$$

$$0s^2 + s + 2 = A(s^2 + 3s) + B(s + 3) + Cs^2$$

Comparing the coefficients, we get

$$A + 0B + C = 0$$

$$3A + B + 0C = 1$$

$$0A + 3B + 0C = 2$$

On solving the equations, we get

$$A = \frac{1}{9}, B = \frac{2}{3}, C = -\frac{1}{9}$$

$$L^{-1} \left\{ \frac{s+2}{s^2(s+3)} \right\} = L^{-1} \left\{ \frac{\frac{1}{9}}{s} + \frac{\frac{2}{3}}{s^2} - \frac{\frac{1}{9}}{s+3} \right\} = \frac{1}{9} + \frac{2}{3}t - \frac{1}{9}e^{-3t}$$

8. Find the Inverse L.T. of $\frac{1}{s^3(s-1)}$

[N14/ElexExtcElectBiomInst/4M]

Solution:

$$\text{Let } \frac{1}{s^3(s-1)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s^3} + \frac{D}{s-1}$$

$$1 = As^2(s-1) + Bs(s-1) + C(s-1) + Ds^3$$

$$1 = A(s^3 - s^2) + B(s^2 - s) + C(s-1) + Ds^3$$

Comparing the coefficients, we get

$$A + D = 0$$

$$-A + B = 0$$

$$-B + C = 0$$

$$-C = 1$$

On solving the equations, we get

$$A = -1, B = -1, C = -1, D = 1$$

$$L^{-1} \left\{ \frac{1}{s^3(s-1)} \right\} = L^{-1} \left\{ -\frac{1}{s} - \frac{1}{s^2} - \frac{1}{s^3} + \frac{1}{s-1} \right\} = -1 - t - \frac{t^2}{2} + e^t$$

9. Find Inverse Laplace Transforms of: $\frac{5s^2-15s-11}{(s+1)(s-2)^2}$

[N13/Chem/6M]

Solution:

$$\text{Let } \frac{5s^2-15s-11}{(s+1)(s-2)^2} = \frac{A}{s+1} + \frac{B}{s-2} + \frac{C}{(s-2)^2}$$

$$5s^2 - 15s - 11 = A(s-2)^2 + B(s+1)(s-2) + C(s+1)$$

$$5s^2 - 15s - 11 = A(s^2 - 4s + 4) + B(s^2 - s - 2) + C(s + 1)$$

Comparing the coefficients, we get



$$A + B + 0C = 5$$

$$-4A - B + C = -15$$

$$4A - 2B + C = -11$$

On solving, we get

$$A = 1, B = 4, C = -7$$

$$\begin{aligned} L^{-1} \left\{ \frac{5s^2 - 15s - 11}{(s+1)(s-2)^2} \right\} &= L^{-1} \left\{ \frac{1}{s+1} + \frac{4}{s-2} - \frac{7}{(s-2)^2} \right\} \\ &= e^{-t} + 4e^{2t} - 7e^{2t} L^{-1} \left\{ \frac{1}{s^2} \right\} \\ &= e^{-t} + 4e^{2t} - 7e^{2t} \cdot t \end{aligned}$$

10. Find Inverse Laplace Transforms of: $\frac{1}{(s-2)(s+2)^2}$ Ans. $\frac{e^{2t}}{16} - \frac{e^{-2t}}{16} - \frac{te^{-2t}}{4}$

11. Find Inverse Laplace Transforms of: $\frac{s}{(s+2)^2(s+1)}$ Ans. $e^{-2t} + 2te^{-2t} - e^{-t}$

12. Find Inverse Laplace Transforms of: $\frac{1}{s(s+4)^2}$ Ans. $\frac{1}{16} - \frac{e^{-4t}}{16} - \frac{te^{-4t}}{4}$

13. Find the inverse Laplace transform of $\frac{s+4}{(s^2-1)(s+1)}$

[N15/ChemBiot/6M][M16/ChemBiot/6M][M18/N18/Chem/6M]

Solution:

$$\text{We have, } L^{-1} \left\{ \frac{s+4}{(s^2-1)(s+1)} \right\} = L^{-1} \left\{ \frac{s+4}{(s-1)(s+1)(s+1)} \right\}$$

$$\text{Let } \frac{s+4}{(s-1)(s+1)^2} = \frac{A}{s-1} + \frac{B}{s+1} + \frac{C}{(s+1)^2}$$

$$s + 4 = A(s+1)^2 + B(s-1)(s+1) + C(s-1)$$

$$s + 4 = A(s^2 + 2s + 1) + B(s^2 - 1) + C(s - 1)$$

On comparing the coefficients, we get

$$A + B = 0$$

$$2A + C = 1$$

$$A - B - C = 4$$

On solving, we get

$$A = \frac{5}{4}, B = -\frac{5}{4}, C = -\frac{3}{2}$$

$$\begin{aligned} L^{-1} \left\{ \frac{s+4}{(s^2-1)(s+1)} \right\} &= L^{-1} \left\{ \frac{\frac{5}{4}}{s-1} - \frac{\frac{5}{4}}{s+1} - \frac{\frac{3}{2}}{(s+1)^2} \right\} \\ &= \frac{5}{4}e^t - \frac{5}{4}e^{-t} - \frac{3}{2}e^{-t} \cdot t \end{aligned}$$

14. Find Inverse Laplace Transforms of: $\frac{3s+1}{(s+1)(s^2+2)}$

[M14/ChemBiot/4M]

Solution:

$$\text{Let } \frac{3s+1}{(s+1)(s^2+2)} = \frac{A}{s+1} + \frac{Bs+C}{s^2+2}$$



$$3s + 1 = A(s^2 + 2) + (Bs + C)(s + 1)$$

$$3s + 1 = A(s^2 + 2) + B(s^2 + s) + C(s + 1)$$

Comparing the coefficients, we get

$$A + B = 0$$

$$B + C = 3$$

$$2A + C = 1$$

On solving the equations, we get

$$A = -\frac{2}{3}, B = \frac{2}{3}, C = \frac{7}{3}$$

$$\begin{aligned} L^{-1} \left\{ \frac{3s+1}{(s+1)(s^2+2)} \right\} &= L^{-1} \left\{ -\frac{\frac{2}{3}}{s+1} + \frac{\frac{2s+7}{3}}{s^2+2} \right\} \\ &= \frac{1}{3} L^{-1} \left\{ -\frac{2}{s+1} + \frac{2s}{s^2+2} + \frac{7}{s^2+2} \right\} \\ &= \frac{1}{3} \left[-2e^{-t} + 2\cos\sqrt{2}t + 7 \cdot \frac{\sin\sqrt{2}t}{\sqrt{2}} \right] \end{aligned}$$

15. Find $L^{-1} \left[\frac{s^2+16s-24}{s^4+20s^2+64} \right]$

[N17/N18/Biot/8M]

Solution:

$$\text{Let } \frac{s^2+16s-24}{(s^2+4)(s^2+16)} = \frac{As+B}{s^2+4} + \frac{Cs+D}{s^2+16}$$

$$s^2 + 16s - 24 = (As + B)(s^2 + 16) + (Cs + D)(s^2 + 4)$$

$$s^2 + 16s - 24 = A(s^3 + 16s) + B(s^2 + 16) + C(s^3 + 4s) + D(s^2 + 4)$$

Comparing the coefficients, we get

$$A + C = 0$$

$$B + D = 1$$

$$16A + 4C = 16$$

$$16B + 4D = -24$$

Solving the equations, we get

$$A = \frac{4}{3}, C = -\frac{4}{3}, B = -\frac{7}{3}, D = \frac{10}{3}$$

Thus,

$$\begin{aligned} L^{-1} \left[\frac{s^2+16s-24}{s^4+20s^2+64} \right] &= L^{-1} \left\{ \frac{\frac{4}{3}s-\frac{7}{3}}{s^2+4} + \frac{\frac{-4}{3}s+\frac{10}{3}}{s^2+16} \right\} \\ &= \frac{1}{3} L^{-1} \left\{ \frac{4s}{s^2+4} - \frac{7}{s^2+4} - \frac{4s}{s^2+16} + \frac{10}{s^2+16} \right\} \\ &= \frac{1}{3} \left[4 \cos 2t - \frac{7 \sin 2t}{2} - 4 \cos 4t + \frac{10 \sin 4t}{4} \right] \\ &= \frac{4}{3} \cos 2t - \frac{7}{6} \sin 2t - \frac{4}{3} \cos 4t + \frac{5}{6} \sin 4t \end{aligned}$$

16. Find Inverse Laplace Transforms of: $\frac{s^2}{(s^2+a^2)(s^2+b^2)}$

[M15/ElexExtcElectBiomInst/4M]

Solution:

For partial fractions, let $s^2 = x$

$$\frac{x}{(x+a^2)(x+b^2)} = \frac{A}{x+a^2} + \frac{B}{x+b^2}$$

$$x = A(x+b^2) + B(x+a^2)$$

Put $x = -a^2$

$$-a^2 = A(-a^2 + b^2)$$

$$A = \frac{a^2}{a^2-b^2}$$

Put $x = -b^2$

$$-b^2 = B(-b^2 + a^2)$$

$$B = -\frac{b^2}{a^2-b^2}$$

$$\begin{aligned} L^{-1} \left\{ \frac{s^2}{(s^2+a^2)(s^2+b^2)} \right\} &= L^{-1} \left\{ \frac{\frac{a^2}{a^2-b^2}}{s^2+a^2} - \frac{\frac{b^2}{a^2-b^2}}{s^2+b^2} \right\} \\ &= \frac{1}{a^2-b^2} L^{-1} \left\{ \frac{a^2}{s^2+a^2} - \frac{b^2}{s^2+b^2} \right\} \\ &= \frac{1}{a^2-b^2} L^{-1} \left\{ a^2 \frac{\sin at}{a} - b^2 \frac{\sin bt}{b} \right\} \\ &= \frac{1}{a^2-b^2} [a \sin at - b \sin bt] \end{aligned}$$

17. Find Inverse Laplace Transforms of: $\frac{s^2+2s+3}{(s^2+2s+5)(s^2+2s+2)}$

[N16/ChemBiot/6M][N17/MTRX/6M]

Solution:

For partial fractions, put $s^2 + 2s = x$

$$\frac{x+3}{(x+5)(x+2)} = \frac{A}{x+5} + \frac{B}{x+2}$$

$$x+3 = A(x+2) + B(x+5)$$

Comparing the coefficients, we get

$$A + B = 1$$

$$2A + 5B = 3$$

On solving, we get

$$A = \frac{2}{3}, B = \frac{1}{3}$$

$$\begin{aligned} L^{-1} \left\{ \frac{s^2+2s+3}{(s^2+2s+5)(s^2+2s+2)} \right\} &= L^{-1} \left\{ \frac{\frac{2}{3}}{s^2+2s+5} + \frac{\frac{1}{3}}{s^2+2s+2} \right\} \\ &= \frac{1}{3} L^{-1} \left\{ \frac{2}{s^2+2s+1+4} + \frac{1}{s^2+2s+1+1} \right\} \\ &= \frac{1}{3} L^{-1} \left\{ \frac{2}{(s+1)^2+4} + \frac{1}{(s+1)^2+1} \right\} \end{aligned}$$

$$= \frac{1}{3} e^{-t} L^{-1} \left\{ \frac{2}{s^2+4} + \frac{1}{s^2+1} \right\}$$

$$= \frac{e^{-t}}{3} [\sin 2t + \sin t]$$

18. Find Inverse L.T. of $\frac{s}{(s^2+4)(s^2+9)}$

[N13/ElexExtcElectBiomInst/4M][N17/AutoMechCivil/6M]

Solution:

$$L^{-1} \left\{ \frac{s}{(s^2+4)(s^2+9)} \right\} = L^{-1} \left\{ s \cdot \frac{1}{(s^2+4)(s^2+9)} \right\}$$

For partial fractions, let $s^2 = x$

$$\frac{1}{(x+4)(x+9)} = \frac{A}{x+4} + \frac{B}{x+9}$$

$$1 = A(x+9) + B(x+4)$$

Comparing the coefficients, we get

$$A + B = 0$$

$$9A + 4B = 1$$

On solving the equations, we get

$$A = \frac{1}{5}, B = -\frac{1}{5}$$

$$L^{-1} \left\{ \frac{s}{(s^2+4)(s^2+9)} \right\} = L^{-1} \left\{ s \cdot \left(\frac{\frac{1}{5}}{s^2+4} - \frac{\frac{1}{5}}{s^2+9} \right) \right\}$$

$$= \frac{1}{5} L^{-1} \left\{ \frac{s}{s^2+4} - \frac{s}{s^2+9} \right\}$$

$$= \frac{1}{5} [\cos 2t - \cos 3t]$$

19. Find Inverse Laplace Transforms of: $\frac{2s}{s^4+4}$

[M15/AutoMechCivil/6M]

Solution:

$$L^{-1} \left\{ \frac{2s}{s^4+4} \right\} = L^{-1} \left\{ \frac{2s}{s^4+4s^2+4-4s^2} \right\}$$

$$= L^{-1} \left\{ \frac{2s}{(s^2+2)^2 - (2s)^2} \right\}$$

$$= L^{-1} \left\{ \frac{2s}{(s^2+2-2s)(s^2+2+2s)} \right\}$$

$$= \frac{1}{2} L^{-1} \left\{ \frac{1}{s^2+2-2s} - \frac{1}{s^2+2+2s} \right\}$$

$$= \frac{1}{2} L^{-1} \left\{ \frac{1}{s^2-2s+1+1} - \frac{1}{s^2+2s+1+1} \right\}$$

$$= \frac{1}{2} L^{-1} \left\{ \frac{1}{(s-1)^2+1} - \frac{1}{(s+1)^2+1} \right\}$$

$$= \frac{1}{2} [e^t \sin t - e^{-t} \sin t]$$

$$= \frac{1}{2} [e^t - e^{-t}] \sin t$$

$$= \sin t \sinh t$$

20. Find the inverse L.T. of $\frac{s+4}{s(s-1)(s^2+4)}$

[M15/ChemBiot/6M]

Solution:

$$\text{Let } \frac{s+4}{s(s-1)(s^2+4)} = \frac{A}{s} + \frac{B}{s-1} + \frac{Cs+D}{s^2+4}$$

$$s+4 = A(s-1)(s^2+4) + Bs(s^2+4) + (Cs+D)s(s-1)$$

$$s+4 = A(s^3+4s-s^2-4) + B(s^3+4s) + C(s^3-s^2) + D(s^2-s)$$

Comparing the coefficients, we get

$$A+B+C=0$$

$$-A-C+D=0$$

$$4A+4B-D=1$$

$$-4A=4$$

On solving the equations, we get

$$A=-1, B=1, C=0, D=-1$$

$$\begin{aligned} L^{-1} \left\{ \frac{s+4}{s(s-1)(s^2+4)} \right\} &= L^{-1} \left\{ -\frac{1}{s} + \frac{1}{s-1} - \frac{1}{s^2+4} \right\} \\ &= -1 + e^t - \frac{\sin 2t}{2} \end{aligned}$$

21. Find Inverse Laplace Transforms of: $\frac{s+29}{(s+4)(s^2+9)}$

[M15/CompIT/4M]

Solution:

$$\text{Let } \frac{s+29}{(s+4)(s^2+9)} = \frac{A}{s+4} + \frac{Bs+C}{s^2+9}$$

$$s+29 = A(s^2+9) + (Bs+C)(s+4)$$

$$s+29 = A(s^2+9) + B(s^2+4s) + C(s+4)$$

Comparing the coefficients, we get

$$A+B=0$$

$$4B+C=1$$

$$9A+4C=29$$

On solving the equations, we get

$$A=1, B=-1, C=5$$

$$\begin{aligned} L^{-1} \left\{ \frac{s+29}{(s+4)(s^2+9)} \right\} &= L^{-1} \left\{ \frac{1}{s+4} + \frac{-s+5}{s^2+9} \right\} \\ &= L^{-1} \left\{ \frac{1}{s+4} - \frac{s}{s^2+9} + \frac{5}{s^2+9} \right\} \\ &= e^{-4t} - \cos 3t + \frac{5}{3} \sin 3t \end{aligned}$$

22. Find Inverse Laplace Transforms of: $\frac{5s^2+8s-1}{(s+3)(s^2+1)}$

[M14/ChemBiot/4M]

Solution:



$$\text{Let } \frac{5s^2+8s-1}{(s+3)(s^2+1)} = \frac{A}{s+3} + \frac{Bs+C}{s^2+1}$$

$$5s^2 + 8s - 1 = A(s^2 + 1) + (Bs + C)(s + 3)$$

$$5s^2 + 8s - 1 = A(s^2 + 1) + B(s^2 + 3s) + C(s + 3)$$

Comparing the coefficients, we get

$$A + B = 5$$

$$3B + C = 8$$

$$A + 3C = -1$$

On solving the equations, we get

$$A = 2, B = 3, C = -1$$

$$\begin{aligned} L^{-1} \left\{ \frac{5s^2+8s-1}{(s+3)(s^2+1)} \right\} &= L^{-1} \left\{ \frac{2}{s+3} + \frac{3s-1}{s^2+1} \right\} \\ &= L^{-1} \left\{ \frac{2}{s+3} + \frac{3s}{s^2+1} - \frac{1}{s^2+1} \right\} \\ &= 2e^{-3t} + 3\cos t - \sin t \end{aligned}$$

23. Find Inverse Laplace Transforms of: $\frac{s^2}{(s^2+1)(s^2+4)}$

$$\text{Ans. } -\frac{1}{3}\sin t + \frac{2}{3}\sin 2t$$

24. Find Inverse Laplace Transforms of: $\frac{s}{(s^2+1)(s^2+4)(s^2+9)}$

[N17/IT/6M]

Solution:

$$L^{-1} \left\{ \frac{s}{(s^2+1)(s^2+4)(s^2+9)} \right\} = L^{-1} \left\{ s \cdot \frac{1}{(s^2+1)(s^2+4)(s^2+9)} \right\}$$

For partial fractions, let $s^2 = x$

$$\frac{1}{(x+1)(x+4)(x+9)} = \frac{A}{x+1} + \frac{B}{x+4} + \frac{C}{x+9}$$

$$1 = A(x+4)(x+9) + B(x+1)(x+9) + C(x+1)(x+4)$$

$$1 = A(x^2 + 13x + 36) + B(x^2 + 10x + 9) + C(x^2 + 5x + 4)$$

Comparing the coefficients, we get

$$A + B + C = 0$$

$$13A + 10B + 5C = 0$$

$$36A + 9B + 4C = 1$$

On solving the equations, we get

$$A = \frac{1}{24}, B = -\frac{1}{15}, C = \frac{1}{40}$$

$$\begin{aligned} L^{-1} \left\{ \frac{s}{(s^2+1)(s^2+4)(s^2+9)} \right\} &= L^{-1} \left\{ s \cdot \left(\frac{\frac{1}{24}}{s^2+1} - \frac{\frac{1}{15}}{s^2+4} + \frac{\frac{1}{40}}{s^2+9} \right) \right\} \\ &= L^{-1} \left\{ \frac{\frac{1}{24}s}{s^2+1} - \frac{\frac{1}{15}s}{s^2+4} + \frac{\frac{1}{40}s}{s^2+9} \right\} \\ &= \frac{\cos t}{24} - \frac{\cos 2t}{15} + \frac{\cos 3t}{40} \end{aligned}$$



25. Find Inverse Laplace Transforms of: $\frac{6s+3}{s^4+5s^2+4}$

$$\text{Ans. } 2\cos t + \sin t - 2\cos 2t - \frac{\sin 2t}{2}$$

26. Find Inverse Laplace Transforms of: $\frac{s^2+2s+3}{(s^2+2s+10)(s^2+2s+17)}$

[M19/IT/6M]

Solution:

For partial fractions, put $s^2 + 2s = x$

$$\frac{x+3}{(x+10)(x+17)} = \frac{A}{x+10} + \frac{B}{x+17}$$

$$x + 3 = A(x + 17) + B(x + 10)$$

Comparing the coefficients, we get

$$A + B = 1$$

$$17A + 10B = 3$$

On solving, we get

$$A = -1, B = 2$$

$$\begin{aligned} L^{-1} \left\{ \frac{s^2+2s+3}{(s^2+2s+10)(s^2+2s+17)} \right\} &= L^{-1} \left\{ \frac{-1}{s^2+2s+10} + \frac{2}{s^2+2s+17} \right\} \\ &= L^{-1} \left\{ \frac{-1}{s^2+2s+1+9} + \frac{2}{s^2+2s+1+16} \right\} \\ &= L^{-1} \left\{ \frac{-1}{(s+1)^2+9} + \frac{2}{(s+1)^2+16} \right\} \\ &= e^{-t} L^{-1} \left\{ \frac{-1}{s^2+9} + \frac{2}{s^2+16} \right\} \\ &= e^{-t} \left[-\frac{\sin 3t}{3} + \frac{2\sin 4t}{4} \right] \\ &= e^{-t} \left[-\frac{\sin 3t}{3} + \frac{\sin 4t}{2} \right] \end{aligned}$$

27. Find Inverse Laplace Transforms of: $\frac{s^2+2s-4}{(s^2+2s+5)(s^2+2s+2)}$

$$\text{Ans. } 3e^{-t} \cdot \frac{1}{2} \cdot \sin 2t - 2e^t \sin t$$

28. Find Inverse Laplace Transforms of: $\frac{2s^2-1}{(s^2+1)(s^2+4)}$

$$\text{Ans. } \frac{3}{2} \sin 2t - \sin t$$

Type IV: Convolution Theorem

1. Find Inverse Laplace Transforms by convolution theorem: $\frac{s^2}{(s^2+a^2)^2}$
 Ans. $\frac{[sinat+atcosat]}{2a}$

2. Find Inverse Laplace Transforms by convolution theorem: $\frac{s^2}{(s^2-a^2)^2}$

[M14/AutoMechCivil/6M]

Solution:

$$\text{Let } F(s) = \frac{s}{s^2-a^2},$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{s}{s^2-a^2}\right\},$$

$$f(t) = \cosh at,$$

$$f(u) = \cosh au,$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(u).g(t-u)du$$

$$\begin{aligned} L^{-1}\left\{\frac{s^2}{(s^2-a^2)^2}\right\} &= \int_0^t \cosh au. \cosh(at-au) du \\ &= \frac{1}{2} \int_0^t 2 \cosh au. \cosh(at-au) du \\ &= \frac{1}{2} \int_0^t \cosh(au+at-au) + \cosh(au-at+au) du \\ &= \frac{1}{2} \int_0^t \cosh at + \cosh(2au-at) du \\ &= \frac{1}{2} \left[u. \cosh at + \frac{\sinh(2au-at)}{2a} \right]_0^t \\ &= \frac{1}{2} \left[t \cosh at + \frac{\sinh at}{2a} - 0 - \frac{\sinh(-at)}{2a} \right] \\ &= \frac{1}{2} \left[t \cosh at + \frac{\sinh at}{a} \right] \\ &= \frac{[sinhat+atcoshat]}{2a} \end{aligned}$$

3. Find Inverse Laplace Transforms by convolution theorem: $\frac{(s+3)^2}{(s^2+6s+13)^2}$
 Ans. $e^{-3t} \cdot \frac{[sin2t+2tcos2t]}{4}$

4. Find Inverse Laplace Transforms by convolution theorem: $\frac{s^2}{(s^2+a^2)(s^2+b^2)}$
 Ans. $\frac{asinat-bsinbt}{a^2-b^2}$

5. By convolution theorem, solve $L^{-1}\left[\frac{s^2}{(s^2+1)(s^2+4)}\right]$

[N18/Biot/6M]

Solution:

$$\text{Let } F(s) = \frac{s}{s^2+1},$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{s}{s^2+1}\right\},$$

$$f(t) = \cos t,$$

$$G(s) = \frac{s}{s^2+4}$$

$$L^{-1}\{G(s)\} = L^{-1}\left\{\frac{s}{s^2+4}\right\}$$

$$g(t) = \cos 2t$$



$$f(u) = \cos u,$$

$$g(t-u) = \cos 2(t-u)$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(u).g(t-u)du$$

$$\begin{aligned} L^{-1}\left\{\frac{s^2}{(s^2+1)(s^2+4)}\right\} &= \int_0^t \cos u. \cos(2t-2u) du \\ &= \frac{1}{2} \int_0^t 2 \cos u. \cos(2t-2u) du \\ &= \frac{1}{2} \int_0^t \cos(u+2t-2u) + \cos(u-2t+2u) du \\ &= \frac{1}{2} \int_0^t \cos(2t-u) + \cos(3u-2t) du \\ &= \frac{1}{2} \left[\frac{\sin(2t-u)}{-1} + \frac{\sin(3u-2t)}{3} \right]_0^t \\ &= \frac{1}{2} \left[\frac{\sin t}{-1} + \frac{\sin t}{3} - \frac{\sin 2t}{-1} - \frac{\sin(-2t)}{3} \right] \\ &= \frac{1}{2} \left[\frac{2 \sin t}{-3} + \frac{4 \sin 2t}{3} \right] \\ &= \frac{1}{3} [2 \sin 2t - \sin t] \end{aligned}$$

6. By convolution theorem, solve $L^{-1}\left[\frac{s^2}{(s^2+3^2)(s^2+5^2)}\right]$

[M19/IT/6M]

Solution:

$$\text{Let } F(s) = \frac{s}{s^2+3^2},$$

$$G(s) = \frac{s}{s^2+5^2}$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{s}{s^2+3^2}\right\},$$

$$L^{-1}\{G(s)\} = L^{-1}\left\{\frac{s}{s^2+5^2}\right\}$$

$$f(t) = \cos 3t,$$

$$g(t) = \cos 5t$$

$$f(u) = \cos 3u,$$

$$g(t-u) = \cos 5(t-u)$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(u).g(t-u)du$$

$$\begin{aligned} L^{-1}\left\{\frac{s^2}{(s^2+3^2)(s^2+5^2)}\right\} &= \int_0^t \cos 3u. \cos(5t-5u) du \\ &= \frac{1}{2} \int_0^t 2 \cos 3u. \cos(5t-5u) du \\ &= \frac{1}{2} \int_0^t \cos(3u+5t-5u) + \cos(3u-5t+5u) du \\ &= \frac{1}{2} \int_0^t \cos(5t-2u) + \cos(8u-5t) du \\ &= \frac{1}{2} \left[\frac{\sin(5t-2u)}{-2} + \frac{\sin(8u-5t)}{8} \right]_0^t \\ &= \frac{1}{2} \left[\frac{\sin 3t}{-2} + \frac{\sin 3t}{8} - \frac{\sin 5t}{-2} - \frac{\sin(-5t)}{8} \right] \\ &= \frac{1}{2} \left[\frac{6 \sin 3t}{-16} + \frac{10 \sin 5t}{16} \right] \\ &= \frac{3 \sin 3t - 5 \sin 5t}{-16} \end{aligned}$$



7. By convolution theorem, solve $L^{-1} \left[\frac{s^2}{(s^2+4)(s^2+25)} \right]$

[M19/Chem/6M]

Solution:

$$\text{Let } F(s) = \frac{s}{s^2+4},$$

$$G(s) = \frac{s}{s^2+25}$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{s}{s^2+4}\right\}, \quad L^{-1}\{G(s)\} = L^{-1}\left\{\frac{s}{s^2+25}\right\}$$

$$f(t) = \cos 2t,$$

$$g(t) = \cos 5t$$

$$f(u) = \cos 2u,$$

$$g(t-u) = \cos 5(t-u)$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(u).g(t-u)du$$

$$\begin{aligned} L^{-1}\left\{\frac{s^2}{(s^2+4)(s^2+25)}\right\} &= \int_0^t \cos 2u. \cos(5t-5u) du \\ &= \frac{1}{2} \int_0^t 2 \cos 2u. \cos(5t-5u) du \\ &= \frac{1}{2} \int_0^t \cos(2u+5t-5u) + \cos(2u-5t+5u) du \\ &= \frac{1}{2} \int_0^t \cos(5t-3u) + \cos(7u-5t) du \\ &= \frac{1}{2} \left[\frac{\sin(5t-3u)}{-3} + \frac{\sin(7u-5t)}{7} \right]_0^t \\ &= \frac{1}{2} \left[\frac{\sin 2t}{-3} + \frac{\sin 2t}{7} - \frac{\sin 5t}{-3} - \frac{\sin(-5t)}{7} \right] \\ &= \frac{1}{2} \left[\frac{4 \sin 2t}{-21} + \frac{10 \sin 5t}{21} \right] \\ &= \frac{1}{21} [5 \sin 5t - 2 \sin 2t] \end{aligned}$$

8. By convolution theorem, solve $L^{-1} \left[\frac{s^2}{(s^2+9)(s^2+16)} \right]$

[M18/Inst/6M]

Solution:

$$\text{Let } F(s) = \frac{s}{s^2+9},$$

$$G(s) = \frac{s}{s^2+16}$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{s}{s^2+9}\right\},$$

$$L^{-1}\{G(s)\} = L^{-1}\left\{\frac{s}{s^2+16}\right\}$$

$$f(t) = \cos 3t,$$

$$g(t) = \cos 4t$$

$$f(u) = \cos 3u,$$

$$g(t-u) = \cos 4(t-u)$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(u).g(t-u)du$$

$$\begin{aligned} L^{-1}\left\{\frac{s^2}{(s^2+9)(s^2+16)}\right\} &= \int_0^t \cos 3u. \cos(4t-4u) du \\ &= \frac{1}{2} \int_0^t 2 \cos 3u. \cos(4t-4u) du \\ &= \frac{1}{2} \int_0^t \cos(3u+4t-4u) + \cos(3u-4t+4u) du \\ &= \frac{1}{2} \int_0^t \cos(4t-u) + \cos(7u-4t) du \end{aligned}$$



$$\begin{aligned}
 &= \frac{1}{2} \left[\frac{\sin(4t-u)}{-1} + \frac{\sin(7u-4t)}{7} \right]_0^t \\
 &= \frac{1}{2} \left[\frac{\sin 3t}{-1} + \frac{\sin 3t}{7} - \frac{\sin 4t}{-1} - \frac{\sin(-4t)}{7} \right] \\
 &= \frac{1}{2} \left[\frac{6\sin 3t}{-7} + \frac{8\sin 4t}{7} \right] \\
 &= \frac{1}{7} [4\sin 4t - 3\sin 3t]
 \end{aligned}$$

9. Find Inverse Laplace Transforms by convolution theorem: $\frac{s^2}{s^4+10s^2+9}$
 Ans. $\frac{1}{8} [3\sin 3t - \sin t]$

10. Using convolution theorem Find $L^{-1} \left[\frac{s^2}{(s^2+4)^2} \right]$
[N15/ElexExtcElectBiomInst/6M][N18/Elex/6M]

Solution:

$$\text{Let } F(s) = \frac{s}{s^2+4},$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{s}{s^2+4}\right\},$$

$$f(t) = \cos 2t,$$

$$f(u) = \cos 2u,$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(u).g(t-u)du$$

$$L^{-1}\left\{\frac{s^2}{(s^2+4)^2}\right\} = \int_0^t \cos 2u. \cos(2t-2u) du$$

$$= \frac{1}{2} \int_0^t 2\cos 2u. \cos(2t-2u) du$$

$$= \frac{1}{2} \int_0^t \cos(2u+2t-2u) + \cos(2u-2t+2u) du$$

$$= \frac{1}{2} \int_0^t \cos 2t + \cos(4u-2t) du$$

$$= \frac{1}{2} \left[u. \cos 2t + \frac{\sin(4u-2t)}{4} \right]_0^t$$

$$= \frac{1}{2} \left[t\cos 2t + \frac{\sin 2t}{4} - 0 - \frac{\sin(-2t)}{4} \right]$$

$$= \frac{1}{2} \left[t\cos 2t + \frac{\sin 2t}{2} \right]$$

$$= \frac{1}{4} [2t\cos 2t + \sin 2t]$$

11. Find Inverse L.T. of: $\frac{(s+2)^2}{(s^2+4s+8)^2}$

[M16/ChemBiot/4M][N16/CompIT/6M][N17/Extc/6M][M18/IT/6M]
[N18/IT/6M][N18/Elect/6M][M19/Elect/6M]

Solution:

$$L^{-1}\left\{\frac{(s+2)^2}{(s^2+4s+8)^2}\right\} = L^{-1}\left\{\frac{(s+2)^2}{(s^2+4s+4+4)^2}\right\}$$



$$= L^{-1} \left\{ \frac{(s+2)^2}{((s+2)^2+4)^2} \right\}$$

$$= e^{-2t} L^{-1} \left\{ \frac{s^2}{(s^2+4)^2} \right\}$$

Now, consider $L^{-1} \left\{ \frac{s^2}{(s^2+4)^2} \right\}$

Let $F(s) = \frac{s}{s^2+4},$

$G(s) = \frac{s}{s^2+4}$

$L^{-1}\{F(s)\} = L^{-1} \left\{ \frac{s}{s^2+4} \right\},$

$L^{-1}\{G(s)\} = L^{-1} \left\{ \frac{s}{s^2+4} \right\}$

$f(t) = \cos 2t,$

$g(t) = \cos 2t$

$f(u) = \cos 2u,$

$g(t-u) = \cos 2(t-u)$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(u).g(t-u)du$$

$$L^{-1} \left\{ \frac{s^2}{(s^2+4)^2} \right\} = \int_0^t \cos 2u. \cos(2t-2u) du$$

$$= \frac{1}{2} \int_0^t 2 \cos 2u. \cos(2t-2u) du$$

$$= \frac{1}{2} \int_0^t \cos(2u+2t-2u) + \cos(2u-2t+2u) du$$

$$= \frac{1}{2} \int_0^t \cos 2t + \cos(4u-2t) du$$

$$= \frac{1}{2} \left[u. \cos 2t + \frac{\sin(4u-2t)}{4} \right]_0^t$$

$$= \frac{1}{2} \left[t \cos 2t + \frac{\sin 2t}{4} - 0 - \frac{\sin(-2t)}{4} \right]$$

$$= \frac{1}{2} \left[t \cos 2t + \frac{\sin 2t}{2} \right]$$

$$= \frac{1}{4} [2t \cos 2t + \sin 2t]$$

Thus,

$$L^{-1} \left\{ \frac{(s+2)^2}{(s^2+4s+8)^2} \right\} = e^{-2t} L^{-1} \left\{ \frac{s^2}{(s^2+4)^2} \right\} = \frac{e^{-2t}}{4} [\sin 2t + 2t \cos 2t]$$

12. Find Inverse Laplace Transforms of: $\frac{(s-1)^2}{(s^2-2s+5)^2}$

[M14/CompIT/6M][M17/CompIT/6M]

Solution:

$$L^{-1} \left\{ \frac{(s-1)^2}{(s^2-2s+5)^2} \right\} = L^{-1} \left\{ \frac{(s-1)^2}{(s^2-2s+1+4)^2} \right\}$$

$$= L^{-1} \left\{ \frac{(s-1)^2}{((s-1)^2+4)^2} \right\}$$

$$= e^t L^{-1} \left\{ \frac{s^2}{(s^2+4)^2} \right\}$$

Now, consider $L^{-1} \left\{ \frac{s^2}{(s^2+4)^2} \right\}$

Let $F(s) = \frac{s}{s^2+4},$

$G(s) = \frac{s}{s^2+4}$

$L^{-1}\{F(s)\} = L^{-1} \left\{ \frac{s}{s^2+4} \right\},$

$L^{-1}\{G(s)\} = L^{-1} \left\{ \frac{s}{s^2+4} \right\}$



$$f(t) = \cos 2t,$$

$$g(t) = \cos 2t$$

$$f(u) = \cos 2u,$$

$$g(t-u) = \cos 2(t-u)$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(u).g(t-u)du$$

$$\begin{aligned} L^{-1}\left\{\frac{s^2}{(s^2+4)^2}\right\} &= \int_0^t \cos 2u. \cos(2t-2u) du \\ &= \frac{1}{2} \int_0^t 2\cos 2u. \cos(2t-2u) du \\ &= \frac{1}{2} \int_0^t \cos(2u+2t-2u) + \cos(2u-2t+2u) du \\ &= \frac{1}{2} \int_0^t \cos 2t + \cos(4u-2t) du \\ &= \frac{1}{2} \left[u. \cos 2t + \frac{\sin(4u-2t)}{4} \right]_0^t \\ &= \frac{1}{2} \left[t\cos 2t + \frac{\sin 2t}{4} - 0 - \frac{\sin(-2t)}{4} \right] \\ &= \frac{1}{2} \left[t\cos 2t + \frac{\sin 2t}{2} \right] \\ &= \frac{1}{4} [2t\cos 2t + \sin 2t] \end{aligned}$$

Thus,

$$L^{-1}\left\{\frac{(s-1)^2}{(s^2-2s+5)^2}\right\} = e^t L^{-1}\left\{\frac{s^2}{(s^2+4)^2}\right\} = \frac{e^t}{4} [\sin 2t + 2t\cos 2t]$$

13. Find Inverse Laplace Transforms of: $\frac{(s+3)^2}{(s^2+6s+5)^2}$ using convolution theorem

[M14/ChemBiot/4M][M16/CompIT/6M]

Solution:

$$\begin{aligned} L^{-1}\left\{\frac{(s+3)^2}{(s^2+6s+5)^2}\right\} &= L^{-1}\left\{\frac{(s+3)^2}{((s+3)^2-4)^2}\right\} \\ &= L^{-1}\left\{\frac{(s+3)^2}{((s+3)^2-4)^2}\right\} \\ &= e^{-3t} L^{-1}\left\{\frac{s^2}{(s^2-4)^2}\right\} \end{aligned}$$

$$\text{Consider, } L^{-1}\left\{\frac{s^2}{(s^2-4)^2}\right\}$$

$$\text{Let } F(s) = \frac{s}{s^2-4},$$

$$G(s) = \frac{s}{s^2-4}$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{s}{s^2-4}\right\},$$

$$L^{-1}\{G(s)\} = L^{-1}\left\{\frac{s}{s^2-4}\right\}$$

$$f(t) = \cosh 2t,$$

$$g(t) = \cosh 2t$$

$$f(u) = \cosh 2u,$$

$$g(t-u) = \cosh 2(t-u)$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(u).g(t-u)du$$

$$\begin{aligned} L^{-1}\left\{\frac{s^2}{(s^2-4)^2}\right\} &= \int_0^t \cosh 2u. \cosh(2t-2u) du \\ &= \frac{1}{2} \int_0^t 2\cosh 2u. \cosh(2t-2u) du \end{aligned}$$



$$\begin{aligned}
 &= \frac{1}{2} \int_0^t \cosh(2u + 2t - 2u) + \cosh(2u - 2t + 2u) du \\
 &= \frac{1}{2} \int_0^t \cosh 2t + \cosh(4u - 2t) du \\
 &= \frac{1}{2} \left[u \cdot \cosh 2t + \frac{\sinh(4u - 2t)}{4} \right]_0^t \\
 &= \frac{1}{2} \left[t \cosh 2t + \frac{\sinh 2t}{4} - 0 - \frac{\sinh(-2t)}{4} \right] \\
 &= \frac{1}{2} \left[t \cosh 2t + \frac{\sinh 2t}{2} \right] \\
 &= \frac{[\sinh 2t + 2t \cosh 2t]}{4}
 \end{aligned}$$

Thus,

$$L^{-1} \left\{ \frac{(s+3)^2}{(s^2+6s+5)^2} \right\} = e^{-3t} L^{-1} \left\{ \frac{s^2}{(s^2-4)^2} \right\} = \frac{e^{-3t}}{4} [2t \cosh 2t + \sinh 2t]$$

14. Find by convolution theorem Inverse L.T. of: $\frac{(s+5)^2}{(s^2+10s+16)^2}$

[N19/IT/6M]

Solution:

$$\begin{aligned}
 L^{-1} \left\{ \frac{(s+5)^2}{(s^2+10s+16)^2} \right\} &= L^{-1} \left\{ \frac{(s+5)^2}{(s^2+10s+25-9)^2} \right\} \\
 &= L^{-1} \left\{ \frac{(s+5)^2}{((s+5)^2-9)^2} \right\} \\
 &= e^{-5t} L^{-1} \left\{ \frac{s^2}{(s^2-9)^2} \right\}
 \end{aligned}$$

Consider, $L^{-1} \left\{ \frac{s^2}{(s^2-9)^2} \right\}$

$$\text{Let } F(s) = \frac{s}{s^2-9},$$

$$G(s) = \frac{s}{s^2-9}$$

$$L^{-1}\{F(s)\} = L^{-1} \left\{ \frac{s}{s^2-9} \right\},$$

$$L^{-1}\{G(s)\} = L^{-1} \left\{ \frac{s}{s^2-9} \right\}$$

$$f(t) = \cosh 3t,$$

$$g(t) = \cosh 3t$$

$$f(u) = \cosh 3u,$$

$$g(t-u) = \cosh 3(t-u)$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(u).g(t-u)du$$

$$\begin{aligned}
 L^{-1} \left\{ \frac{s^2}{(s^2-9)^2} \right\} &= \int_0^t \cosh 3u. \cosh(3t-3u) du \\
 &= \frac{1}{2} \int_0^t 2 \cosh 3u. \cosh(3t-3u) du \\
 &= \frac{1}{2} \int_0^t \cosh(3u+3t-3u) + \cosh(3u-3t+3u) du \\
 &= \frac{1}{2} \int_0^t \cosh 3t + \cosh(6u-3t) du \\
 &= \frac{1}{2} \left[u \cdot \cosh 3t + \frac{\sinh(6u-3t)}{6} \right]_0^t \\
 &= \frac{1}{2} \left[t \cosh 3t + \frac{\sinh 3t}{6} - 0 - \frac{\sinh(-3t)}{6} \right] \\
 &= \frac{1}{2} \left[t \cosh 3t + \frac{\sinh 3t}{3} \right]
 \end{aligned}$$



Thus,

$$L^{-1}\left\{\frac{(s+5)^2}{(s^2+10s+16)^2}\right\} = e^{-5t}L^{-1}\left\{\frac{s^2}{(s^2-9)^2}\right\} = \frac{e^{-5t}}{2}\left[t\cosh 3t + \frac{\sinh 3t}{3}\right]$$

15. Find Inverse Laplace Transforms by convolution theorem: $\frac{s}{(s^2+a^2)^2}$

[M19/Extc/6M]

Solution:

$$\text{Let } F(s) = \frac{1}{s^2+a^2},$$

$$G(s) = \frac{s}{s^2+a^2}$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{1}{s^2+a^2}\right\},$$

$$L^{-1}\{G(s)\} = L^{-1}\left\{\frac{s}{s^2+a^2}\right\}$$

$$f(t) = \frac{\sin at}{a},$$

$$g(t) = \cos at$$

$$f(u) = \frac{\sin au}{a},$$

$$g(t-u) = \cos a(t-u)$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(u).g(t-u)du$$

$$\begin{aligned} L^{-1}\left\{\frac{s}{(s^2+a^2)^2}\right\} &= \int_0^t \frac{\sin au}{a} \cdot \cos(at-au) du \\ &= \frac{1}{2a} \int_0^t 2\sin au \cdot \cos(at-au) du \\ &= \frac{1}{2a} \int_0^t \sin(au+at-au) + \sin(au-at+au) du \\ &= \frac{1}{2a} \int_0^t \sin at + \sin(2au-at) du \\ &= \frac{1}{2a} \left[u \cdot \sin at - \frac{\cos(2au-at)}{2a} \right]_0^t \\ &= \frac{1}{2a} \left[t \sin at - \frac{\cos at}{2a} - 0 + \frac{\cos(-at)}{2a} \right] \\ &= \frac{1}{2a} [t \sin at] \end{aligned}$$

16. Find Inverse Laplace Transforms by convolution theorem: $\frac{s}{(s^2+a^2)(s^2+b^2)}$

[N16/ElexExtcElectBiomInst/6M]

Solution:

$$\text{Let } F(s) = \frac{1}{s^2+a^2},$$

$$G(s) = \frac{s}{s^2+b^2}$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{1}{s^2+a^2}\right\},$$

$$L^{-1}\{G(s)\} = L^{-1}\left\{\frac{s}{s^2+b^2}\right\}$$

$$f(t) = \frac{\sin at}{a},$$

$$g(t) = \cos bt$$

$$f(u) = \frac{\sin au}{a},$$

$$g(t-u) = \cos b(t-u)$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(u).g(t-u)du$$

$$L^{-1}\left\{\frac{s}{(s^2+a^2)(s^2+b^2)}\right\} = \int_0^t \frac{\sin au}{a} \cdot \cos(bt-bu) du$$

$$\begin{aligned}
 &= \frac{1}{2a} \int_0^t 2 \sin au \cdot \cos(bt - bu) du \\
 &= \frac{1}{2a} \int_0^t \sin(au + bt - bu) + \sin(au - bt + bu) du \\
 &= \frac{1}{2a} \left[-\frac{\cos(au - bu + bt)}{a-b} - \frac{\cos(au + bu - bt)}{a+b} \right]_0^t \\
 &= \frac{1}{2a} \left[-\frac{\cos at}{a-b} - \frac{\cos at}{a+b} + \frac{\cos bt}{a-b} + \frac{\cos bt}{a+b} \right] \\
 &= \frac{1}{2a} \left[\frac{-2a \cos at}{a^2 - b^2} + \frac{2a \cos bt}{a^2 - b^2} \right] \\
 &= \frac{\cos bt - \cos at}{a^2 - b^2}
 \end{aligned}$$

17. Find Inverse Laplace Transforms by convolution theorem: $\frac{s}{(s^2+9)^2}$

[N17/Elex/6M]

Solution:

$$\text{Let } F(s) = \frac{1}{s^2+9},$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{1}{s^2+9}\right\},$$

$$f(t) = \frac{\sin 3t}{3},$$

$$f(u) = \frac{\sin 3u}{3},$$

$$G(s) = \frac{s}{s^2+9}$$

$$L^{-1}\{G(s)\} = L^{-1}\left\{\frac{s}{s^2+9}\right\}$$

$$g(t) = \cos 3t$$

$$g(t-u) = \cos 3(t-u)$$

By convolution theorem,

$$L^{-1}\{F(s) \cdot G(s)\} = \int_0^t f(u) \cdot g(t-u) du$$

$$\begin{aligned}
 L^{-1}\left\{\frac{s}{(s^2+9)^2}\right\} &= \int_0^t \frac{\sin 3u}{3} \cdot \cos(3t-3u) du \\
 &= \frac{1}{6} \int_0^t 2 \sin 3u \cdot \cos(3t-3u) du \\
 &= \frac{1}{6} \int_0^t \sin(3u+3t-3u) + \sin(3u-3t+3u) du \\
 &= \frac{1}{6} \int_0^t \sin 3t + \sin(6u-3t) du \\
 &= \frac{1}{6} \left[u \cdot \sin 3t - \frac{\cos(6u-3t)}{6} \right]_0^t \\
 &= \frac{1}{6} \left[t \sin 3t - \frac{\cos 3t}{6} - 0 + \frac{\cos(-3t)}{6} \right] = \frac{1}{6} [t \sin 3t]
 \end{aligned}$$

18. Find Inverse Laplace Transforms by convolution theorem: $\frac{s}{s^4+8s^2+16}$

[M15/CompIT/6M][N17/M18/N18/Chem/8M][M18/Extc/6M]

[N18/AutoMechCivil/6M][N18/Extc/6M]

Solution:

$$L^{-1}\left\{\frac{s}{s^4+8s^2+16}\right\} = L^{-1}\left\{\frac{s}{(s^2+4)^2}\right\}$$

$$\text{Let } F(s) = \frac{1}{s^2+4},$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{1}{s^2+4}\right\},$$

$$G(s) = \frac{s}{s^2+4}$$

$$L^{-1}\{G(s)\} = L^{-1}\left\{\frac{s}{s^2+4}\right\}$$



$$f(t) = \frac{\sin 2t}{2},$$

$$g(t) = \cos 2t$$

$$f(u) = \frac{\sin 2u}{2},$$

$$g(t-u) = \cos 2(t-u)$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(u).g(t-u)du$$

$$\begin{aligned} L^{-1}\left\{\frac{s}{(s^2+4)^2}\right\} &= \int_0^t \frac{\sin 2u}{2} \cdot \cos(2t-2u) du \\ &= \frac{1}{4} \int_0^t 2 \sin 2u \cdot \cos(2t-2u) du \\ &= \frac{1}{4} \int_0^t \sin(2u+2t-2u) + \sin(2u-2t+2u) du \\ &= \frac{1}{4} \int_0^t \sin 2t + \sin(4u-2t) du \\ &= \frac{1}{4} \left[u \cdot \sin 2t - \frac{\cos(4u-2t)}{4} \right]_0^t \\ &= \frac{1}{4} \left[t \sin 2t - \frac{\cos 2t}{4} - 0 + \frac{\cos(-2t)}{4} \right] \\ &= \frac{1}{4} [t \sin 2t] \end{aligned}$$

19. Find Inverse Laplace Transforms by convolution theorem: $\frac{s}{(s^2-a^2)^2}$

[M15/AutoMechCivil/6M][M17/ElexExtcElectBiomInst/6M]

Solution:

$$\text{Let } F(s) = \frac{1}{s^2-a^2},$$

$$G(s) = \frac{s}{s^2-a^2}$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{1}{s^2-a^2}\right\},$$

$$L^{-1}\{G(s)\} = L^{-1}\left\{\frac{s}{s^2-a^2}\right\}$$

$$f(t) = \frac{\sinh at}{a},$$

$$g(t) = \cosh at$$

$$f(u) = \frac{\sinh au}{a},$$

$$g(t-u) = \cosh a(t-u)$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(u).g(t-u)du$$

$$\begin{aligned} L^{-1}\left\{\frac{s}{(s^2-a^2)^2}\right\} &= \int_0^t \frac{\sinh au}{a} \cdot \cosh(at-au) du \\ &= \frac{1}{2a} \int_0^t 2 \sinh au \cdot \cosh(at-au) du \\ &= \frac{1}{2a} \int_0^t \sinh(au+at-au) + \sinh(au-at+au) du \\ &= \frac{1}{2a} \int_0^t \sinh at + \sinh(2au-at) du \\ &= \frac{1}{2a} \left[u \cdot \sinh at + \frac{\cosh(2au-at)}{2a} \right]_0^t \\ &= \frac{1}{2a} \left[t \sinh at + \frac{\cosh at}{2a} - 0 - \frac{\cosh(-at)}{2a} \right] \\ &= \frac{1}{2a} [t \sinh at] \\ &= \frac{t \sinh at}{2a} \end{aligned}$$



20. Find Inverse Laplace Transforms by convolution theorem: $\frac{s+2}{(s^2+4s+5)^2}$

$$\text{Ans. } \frac{e^{-2t}}{2} t \sin t$$

21. Find Inverse Laplace Transforms by convolution theorem: $\frac{s+2}{(s^2+4s+8)^2}$

$$\text{Ans. } e^{-2t} \cdot \frac{t \sin 2t}{4}$$

22. Using Convolution find $L^{-1} \left[\frac{s}{(s^2+1)(s^2+4)} \right]$

[M16/ElexExtcElectBiomInst/6M][M17/AutoMechCivil/6M]

[N17/MTRX/6M]

Solution:

$$\text{Let } F(s) = \frac{1}{s^2+1},$$

$$G(s) = \frac{s}{s^2+4}$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{1}{s^2+1}\right\},$$

$$L^{-1}\{G(s)\} = L^{-1}\left\{\frac{s}{s^2+4}\right\}$$

$$f(t) = \sin t,$$

$$g(t) = \cos 2t$$

$$f(u) = \sin u,$$

$$g(t-u) = \cos 2(t-u)$$

By convolution theorem,

$$L^{-1}\{F(s) \cdot G(s)\} = \int_0^t f(u) \cdot g(t-u) du$$

$$\begin{aligned} L^{-1}\left\{\frac{s}{(s^2+1)(s^2+4)}\right\} &= \int_0^t \sin u \cdot \cos(2t-2u) du \\ &= \frac{1}{2} \int_0^t 2 \sin u \cdot \cos(2t-2u) du \\ &= \frac{1}{2} \int_0^t \sin(u+2t-2u) + \sin(u-2t+2u) du \\ &= \frac{1}{2} \int_0^t \sin(2t-u) + \sin(3u-2t) du \\ &= \frac{1}{2} \left[-\frac{\cos(2t-u)}{-1} - \frac{\cos(3u-2t)}{3} \right]_0^t \\ &= \frac{1}{2} \left[\cos t - \frac{\cos t}{3} - \cos 2t + \frac{\cos 2t}{3} \right] \\ &= \frac{1}{2} \left[\frac{2 \cos t}{3} - \frac{2 \cos 2t}{3} \right] \\ &= \frac{1}{3} [\cos t - \cos 2t] \end{aligned}$$

23. Using Convolution find $L^{-1} \left[\frac{s}{(s^2+4)(s^2+9)} \right]$

[N17/Inst/6M][M18/Elect/6M][M19/Biom/6M]

Solution:

$$\text{Let } F(s) = \frac{1}{s^2+9},$$

$$G(s) = \frac{s}{s^2+4}$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{1}{s^2+9}\right\},$$

$$L^{-1}\{G(s)\} = L^{-1}\left\{\frac{s}{s^2+4}\right\}$$

$$f(t) = \frac{\sin 3t}{3},$$

$$g(t) = \cos 2t$$

$$f(u) = \frac{\sin 3u}{3},$$

$$g(t-u) = \cos 2(t-u)$$

By convolution theorem,



$$\begin{aligned}
 L^{-1}\{F(s).G(s)\} &= \int_0^t f(u).g(t-u)du \\
 L^{-1}\left\{\frac{s}{(s^2+9)(s^2+4)}\right\} &= \int_0^t \frac{\sin 3u}{3} \cdot \cos(2t-2u) du \\
 &= \frac{1}{6} \int_0^t 2\sin 3u \cdot \cos(2t-2u) du \\
 &= \frac{1}{6} \int_0^t \sin(3u+2t-2u) + \sin(3u-2t+2u) du \\
 &= \frac{1}{6} \int_0^t \sin(2t+u) + \sin(5u-2t) du \\
 &= \frac{1}{6} \left[-\frac{\cos(2t+u)}{1} - \frac{\cos(5u-2t)}{5} \right]_0^t \\
 &= \frac{1}{6} \left[-\cos 3t - \frac{\cos 3t}{5} + \cos 2t + \frac{\cos(-2t)}{5} \right] \\
 &= \frac{1}{6} \left[\frac{-6\cos 3t}{5} + \frac{6\cos 2t}{5} \right] \\
 &= \frac{1}{5} [\cos 2t - \cos 3t]
 \end{aligned}$$

24. Using Convolution find $L^{-1}\left[\frac{s}{(s^2+4)(s^2+16)}\right]$

[M18/N18/Biom/6M]

Solution:

$$\text{Let } F(s) = \frac{1}{s^2+16},$$

$$G(s) = \frac{s}{s^2+4}$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{1}{s^2+16}\right\},$$

$$L^{-1}\{G(s)\} = L^{-1}\left\{\frac{s}{s^2+4}\right\}$$

$$f(t) = \frac{\sin 4t}{4},$$

$$g(t) = \cos 2t$$

$$f(u) = \frac{\sin 4u}{4},$$

$$g(t-u) = \cos 2(t-u)$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(u).g(t-u)du$$

$$\begin{aligned}
 L^{-1}\left\{\frac{s}{(s^2+16)(s^2+4)}\right\} &= \int_0^t \frac{\sin 4u}{4} \cdot \cos(2t-2u) du \\
 &= \frac{1}{8} \int_0^t 2\sin 4u \cdot \cos(2t-2u) du \\
 &= \frac{1}{8} \int_0^t \sin(4u+2t-2u) + \sin(4u-2t+2u) du \\
 &= \frac{1}{8} \int_0^t \sin(2t+2u) + \sin(6u-2t) du \\
 &= \frac{1}{8} \left[-\frac{\cos(2t+2u)}{2} - \frac{\cos(6u-2t)}{6} \right]_0^t \\
 &= \frac{1}{8} \left[-\frac{\cos 4t}{2} - \frac{\cos 4t}{6} + \frac{\cos 2t}{2} + \frac{\cos(-2t)}{6} \right] \\
 &= \frac{1}{8} \left[\frac{-8\cos 4t}{12} + \frac{8\cos 2t}{12} \right] \\
 &= \frac{1}{12} [\cos 2t - \cos 4t]
 \end{aligned}$$

25. Find Inverse Laplace Transforms by convolution theorem: $\frac{1}{(s^2+1)^2}$
 Ans. $\frac{[sint-tcost]}{2}$

26. Find the inverse Laplace Transform by using convolution theorem

$$\frac{1}{(s^2+a^2)(s^2+b^2)}$$

[M18/Comp/6M]

Solution:

$$L^{-1}\left\{\frac{1}{(s^2+a^2)(s^2+b^2)}\right\}$$

$$\text{Let } F(s) = \frac{1}{s^2+a^2},$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{1}{s^2+a^2}\right\},$$

$$f(t) = \frac{\sin at}{a},$$

$$f(u) = \frac{\sin au}{a},$$

$$G(s) = \frac{1}{s^2+b^2}$$

$$L^{-1}\{G(s)\} = L^{-1}\left\{\frac{1}{s^2+b^2}\right\}$$

$$g(t) = \frac{\sin bt}{b}$$

$$g(t-u) = \frac{\sin b(t-u)}{b}$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(u).g(t-u)du$$

$$\begin{aligned} L^{-1}\left\{\frac{1}{(s^2+a^2)(s^2+b^2)}\right\} &= \int_0^t \frac{\sin au}{a} \cdot \frac{\sin b(t-u)}{b} du \\ &= \frac{1}{2ab} \int_0^t 2 \sin au \sin(b t - bu) du \\ &= \frac{1}{2ab} \int_0^t [\cos(au - bt + bu) - \cos(au + bt - bu)] du \\ &= \frac{1}{2ab} \left[\frac{\sin(au - bt + bu)}{a+b} - \frac{\sin(au + bt - bu)}{a-b} \right]_0^t \\ &= \frac{1}{2ab} \left[\frac{\sin at}{a+b} - \frac{\sin at}{a-b} - \frac{\sin(-bt)}{a+b} + \frac{\sin(bt)}{a-b} \right] \\ &= \frac{1}{2ab} \left[\frac{-2b \sin at}{a^2-b^2} + \frac{2a \sin bt}{a^2-b^2} \right] \\ &= \frac{1}{a^2-b^2} \left[\frac{\sin bt}{b} - \frac{\sin at}{a} \right] \end{aligned}$$

27. Find Inverse Laplace Transforms by convolution theorem

$$\frac{s^2+2s+3}{(s^2+2s+5)(s^2+2s+2)}$$

[N13/CompIT/6M]

Solution:

$$\begin{aligned} L^{-1}\left\{\frac{s^2+2s+3}{(s^2+2s+5)(s^2+2s+2)}\right\} &= L^{-1}\left\{\frac{s^2+2s+1+2}{(s^2+2s+1+4)(s^2+2s+1+1)}\right\} \\ &= L^{-1}\left\{\frac{(s+1)^2+2}{((s+1)^2+4)((s+1)^2+1)}\right\} \\ &= e^{-t} L^{-1}\left\{\frac{s^2+2}{(s^2+4)(s^2+1)}\right\} \\ &= e^{-t} L^{-1}\left\{\frac{s^2}{(s^2+4)(s^2+1)} + \frac{2}{(s^2+4)(s^2+1)}\right\} \end{aligned}$$



Consider, $L^{-1}\left\{\frac{s^2}{(s^2+4)(s^2+1)}\right\}$

$$\text{Let } F(s) = \frac{s}{s^2+4},$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{s}{s^2+4}\right\},$$

$$f(t) = \cos 2t,$$

$$f(u) = \cos 2u,$$

Consider, $L^{-1}\left\{\frac{2}{(s^2+4)(s^2+1)}\right\}$

$$\text{Let } F(s) = \frac{2}{s^2+4},$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{2}{s^2+4}\right\},$$

$$f(t) = \sin 2t,$$

$$f(u) = \sin 2u,$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(u).g(t-u)du$$

$$\begin{aligned} L^{-1}\left\{\frac{s^2+2s+3}{(s^2+2s+5)(s^2+2s+2)}\right\} &= e^{-t} \int_0^t \cos 2u \cos(t-u) + \sin 2u \sin(t-u) du \\ &= e^{-t} \int_0^t \cos[2u - (t-u)] du \\ &= e^{-t} \int_0^t \cos(3u - t) du \\ &= e^{-t} \left[\frac{\sin(3u-t)}{3} \right]_0^t \\ &= e^{-t} \left[\frac{\sin 2t}{3} - \frac{\sin(-t)}{3} \right] = \frac{e^{-t}}{3} [\sin 2t + \sin t] \end{aligned}$$

28. Find the inverse Laplace Transform by using convolution theorem

$$\frac{1}{(s^2+9)(s^2+4)}$$

[M18/AutoMechCivil/6M]

Solution:

$$L^{-1}\left\{\frac{1}{(s^2+9)(s^2+4)}\right\}$$

$$\text{Let } F(s) = \frac{1}{s^2+9},$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{1}{s^2+9}\right\},$$

$$f(t) = \frac{\sin 3t}{3},$$

$$f(u) = \frac{\sin 3u}{3},$$

$$G(s) = \frac{1}{s^2+4}$$

$$L^{-1}\{G(s)\} = L^{-1}\left\{\frac{1}{s^2+4}\right\}$$

$$g(t) = \frac{\sin 2t}{2}$$

$$g(t-u) = \frac{\sin 2(t-u)}{2}$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(u).g(t-u)du$$

$$\begin{aligned} L^{-1}\left\{\frac{1}{(s^2+9)(s^2+4)}\right\} &= \int_0^t \frac{\sin 3u}{3} \cdot \frac{\sin(2t-2u)}{2} du \\ &= \frac{1}{12} \int_0^t 2 \sin 3u \sin(2t-2u) du \end{aligned}$$



$$\begin{aligned}
 &= \frac{1}{12} \int_0^t \cos(3u - 2t + 2u) - \cos(3u + 2t - 2u) du \\
 &= \frac{1}{12} \left[\frac{\sin(5u-2t)}{5} - \frac{\sin(u+2t)}{1} \right]_0^t \\
 &= \frac{1}{12} \left[\frac{\sin 3t}{5} - \sin 3t - \frac{\sin(-2t)}{5} + \sin 2t \right] \\
 &= \frac{1}{12} \left[\frac{-4\sin 3t}{5} + \frac{6\sin 2t}{5} \right] \\
 &= \frac{1}{5} \left[\frac{\sin 2t}{2} - \frac{\sin 3t}{3} \right]
 \end{aligned}$$

29. Find the inverse Laplace Transform by using convolution theorem

$$\frac{1}{(s^2+9)(s^2+1)}$$

[M19/Comp/6M]

Solution:

$$L^{-1} \left\{ \frac{1}{(s^2+9)(s^2+1)} \right\}$$

$$\text{Let } F(s) = \frac{1}{s^2+9},$$

$$L^{-1}\{F(s)\} = L^{-1} \left\{ \frac{1}{s^2+9} \right\},$$

$$f(t) = \frac{\sin 3t}{3},$$

$$f(u) = \frac{\sin 3u}{3},$$

$$G(s) = \frac{1}{s^2+1}$$

$$L^{-1}\{G(s)\} = L^{-1} \left\{ \frac{1}{s^2+1} \right\}$$

$$g(t) = \sin t$$

$$g(t-u) = \sin(t-u)$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(u).g(t-u)du$$

$$\begin{aligned}
 L^{-1} \left\{ \frac{1}{(s^2+9)(s^2+4)} \right\} &= \int_0^t \frac{\sin 3u}{3} \cdot \sin(t-u) du \\
 &= \frac{1}{6} \int_0^t 2\sin 3u \sin(t-u) du \\
 &= \frac{1}{6} \int_0^t \cos(3u - t + u) - \cos(3u + t - u) du \\
 &= \frac{1}{6} \left[\frac{\sin(4u-t)}{4} - \frac{\sin(2u+t)}{2} \right]_0^t \\
 &= \frac{1}{6} \left[\frac{\sin 3t}{4} - \frac{\sin 3t}{2} - \frac{\sin(-t)}{4} + \frac{\sin t}{2} \right] \\
 &= \frac{1}{6} \left[\frac{-2\sin 3t}{8} + \frac{6\sin 2t}{8} \right] \\
 &= \frac{1}{8} \left[\sin t - \frac{\sin 3t}{3} \right]
 \end{aligned}$$

30. Find Inverse Laplace Transforms by convolution theorem: $\frac{1}{(s^2+4s+13)^2}$

[N13/AutoMechCivil/6M][M18/Elex/6M][M19/Elex/6M]

[N19/Chem/6M]

Solution:

$$L^{-1} \left\{ \frac{1}{(s^2+4s+13)^2} \right\} = L^{-1} \left\{ \frac{1}{(s^2+4s+4+9)^2} \right\}$$



$$= L^{-1} \left\{ \frac{1}{((s+2)^2+9)^2} \right\}$$

$$= e^{-2t} L^{-1} \left\{ \frac{1}{(s^2+9)^2} \right\}$$

Now, consider, $L^{-1} \left\{ \frac{1}{(s^2+9)^2} \right\}$

$$\text{Let } F(s) = \frac{1}{s^2+9},$$

$$G(s) = \frac{1}{s^2+9}$$

$$L^{-1}\{F(s)\} = L^{-1} \left\{ \frac{1}{s^2+9} \right\},$$

$$L^{-1}\{G(s)\} = L^{-1} \left\{ \frac{1}{s^2+9} \right\}$$

$$f(t) = \frac{\sin 3t}{3},$$

$$g(t) = \frac{\sin 3t}{3}$$

$$f(u) = \frac{\sin 3u}{3},$$

$$g(t-u) = \frac{\sin 3(t-u)}{3}$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(u).g(t-u)du$$

$$L^{-1} \left\{ \frac{1}{(s^2+9)^2} \right\} = \int_0^t \frac{\sin 3u}{3} \cdot \frac{\sin 3(t-u)}{3} du$$

$$= \frac{1}{9} \cdot \frac{1}{2} \int_0^t 2 \sin 3u \sin(3t-3u) du$$

$$= \frac{1}{18} \int_0^t \cos(3u-3t+3u) - \cos(3u+3t-3u) du$$

$$= \frac{1}{18} \int_0^t \cos(6u-3t) - \cos(3t) du$$

$$= \frac{1}{18} \left[\frac{\sin(6u-3t)}{6} - u \cdot \cos 3t \right]_0^t$$

$$= \frac{1}{18} \left[\frac{\sin 3t}{6} - t \cos 3t - \frac{\sin(-3t)}{6} + 0 \right]$$

$$= \frac{1}{18} \left[\frac{\sin 3t}{3} - t \cos 3t \right]$$

$$L^{-1} \left\{ \frac{1}{(s^2+4s+13)^2} \right\} = e^{-2t} L^{-1} \left\{ \frac{1}{(s^2+9)^2} \right\} = \frac{e^{-2t}}{18} \left[\frac{\sin 3t}{3} - t \cos 3t \right]$$

31. Find Inverse Laplace Transforms by convolution theorem: $\frac{1}{s^2(s+1)^2}$

[N14/AutoMechCivil/6M]

Solution:

$$\text{Let } F(s) = \frac{1}{s^2},$$

$$G(s) = \frac{1}{(s+1)^2}$$

$$L^{-1}\{F(s)\} = L^{-1} \left\{ \frac{1}{s^2} \right\},$$

$$L^{-1}\{G(s)\} = L^{-1} \left\{ \frac{1}{(s+1)^2} \right\}$$

$$f(t) = t,$$

$$g(t) = e^{-t}t$$

$$f(t-u) = t-u,$$

$$g(u) = e^{-u}u$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(t-u).g(u)du$$

$$L^{-1} \left\{ \frac{1}{s^2(s+1)^2} \right\} = \int_0^t (t-u)e^{-u}.udu = \int_0^t (ut-u^2)e^{-u}du$$

$$= \left[(ut-u^2) \left(\frac{e^{-u}}{-1} \right) - (t-2u) \left(\frac{e^{-u}}{1} \right) + (-2) \left(\frac{e^{-u}}{-1} \right) \right]_0^t$$



$$= te^{-t} + 2e^{-t} + t - 2$$

32. Find Inverse Laplace Transforms using convolution theorem: $\frac{1}{(s-a)(s+a)^2}$

[N17/Comp/6M][N19/Extc/6M]

Solution:

$$\text{Let } F(s) = \frac{1}{(s+a)^2},$$

$$G(s) = \frac{1}{s-a}$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{1}{(s+a)^2}\right\},$$

$$L^{-1}\{G(s)\} = L^{-1}\left\{\frac{1}{s-a}\right\}$$

$$f(t) = e^{-at}t,$$

$$g(t) = e^{at}$$

$$f(u) = e^{-au}u,$$

$$g(t-u) = e^{a(t-u)}$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(u).g(t-u)du$$

$$L^{-1}\left\{\frac{1}{(s+a)^2(s-a)}\right\} = \int_0^t e^{-au}.u.e^{at-au}du$$

$$= e^{at} \int_0^t ue^{-2au}du$$

$$= e^{at} \left[u \left(\frac{e^{-2au}}{-2a} \right) - (1) \left(\frac{e^{-2au}}{4a^2} \right) \right]_0^t$$

$$= e^{at} \left[-\frac{te^{-2at}}{2a} - \frac{e^{-2at}}{4a^2} - 0 + \frac{e^0}{4a^2} \right]$$

$$= e^{at} \left[\frac{1}{4a^2} - \frac{e^{-2at}}{4a^2} - \frac{te^{-at}}{2a} \right]$$

33. Find Inverse Laplace Transforms by convolution theorem: $\frac{1}{(s-4)^4(s+3)}$

[N13/Biot/6M]

Solution:

$$\text{Let } F(s) = \frac{1}{(s-4)^4},$$

$$G(s) = \frac{1}{s+3}$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{1}{(s-4)^4}\right\},$$

$$L^{-1}\{G(s)\} = L^{-1}\left\{\frac{1}{s+3}\right\}$$

$$f(t) = e^{4t} \frac{t^3}{3!},$$

$$g(t) = e^{-3t}$$

$$f(u) = \frac{e^{4u}u^3}{6},$$

$$g(t-u) = e^{-3(t-u)}$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(u).g(t-u)du$$

$$L^{-1}\left\{\frac{1}{(s-4)^4(s+3)}\right\} = \int_0^t \frac{e^{4u}u^3}{6}.e^{-3t+3u}du$$

$$= \frac{e^{-3t}}{6} \int_0^t u^3 e^{7u} du$$

$$= \frac{e^{-3t}}{6} \left[u^3 \left(\frac{e^{7u}}{7} \right) - (3u^2) \left(\frac{e^{7u}}{49} \right) + (6u) \left(\frac{e^{7u}}{343} \right) - 6 \left(\frac{e^{7u}}{2401} \right) \right]_0^t$$

$$= \frac{e^{-3t}}{6} \left[\frac{t^3 e^{7t}}{7} - \frac{3t^2 e^{7t}}{49} + \frac{6t e^{7t}}{343} - \frac{6e^{7t}}{2401} + \frac{6}{2401} \right]$$



34. Find Inverse Laplace Transforms of: $\frac{1}{s^2(s+a)^2}$ by convolution theorem

[N16/ChemBiot/6M]

Solution:

$$\text{Let } F(s) = \frac{1}{s^2},$$

$$G(s) = \frac{1}{(s+a)^2}$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{1}{s^2}\right\},$$

$$L^{-1}\{G(s)\} = L^{-1}\left\{\frac{1}{(s+a)^2}\right\}$$

$$f(t) = t,$$

$$g(t) = e^{-at}t$$

$$f(t-u) = t-u,$$

$$g(u) = e^{-au}u$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(t-u).g(u)du$$

$$\begin{aligned} L^{-1}\left\{\frac{1}{s^2(s+a)^2}\right\} &= \int_0^t (t-u)e^{-au}.u du = \int_0^t (ut - u^2)e^{-au} du \\ &= \left[(ut - u^2) \left(\frac{e^{-au}}{-a}\right) - (t - 2u) \left(\frac{e^{-au}}{a^2}\right) + (-2) \left(\frac{e^{-au}}{-a^3}\right) \right]_0^t \\ &= \frac{te^{-at}}{a^2} + \frac{2e^{-at}}{a^3} + \frac{t}{a^2} - \frac{2}{a^3} \end{aligned}$$

35. Using Convolution Theorem, find the inverse Laplace transform of $F(s) = \frac{1}{s^2(s+5)^2}$

[M19/AutoMechCivil/6M]

Solution:

$$\text{Let } F(s) = \frac{1}{s^2},$$

$$G(s) = \frac{1}{(s+5)^2}$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{1}{s^2}\right\},$$

$$L^{-1}\{G(s)\} = L^{-1}\left\{\frac{1}{(s+5)^2}\right\}$$

$$f(t) = t,$$

$$g(t) = e^{-5t}t$$

$$f(t-u) = t-u,$$

$$g(u) = e^{-5u}u$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(t-u).g(u)du$$

$$\begin{aligned} L^{-1}\left\{\frac{1}{s^2(s+5)^2}\right\} &= \int_0^t (t-u)e^{-5u}.u du = \int_0^t (ut - u^2)e^{-5u} du \\ &= \left[(ut - u^2) \left(\frac{e^{-5u}}{-5}\right) - (t - 2u) \left(\frac{e^{-5u}}{25}\right) + (-2) \left(\frac{e^{-5u}}{-125}\right) \right]_0^t \\ &= 0 + t \left(\frac{e^{-5t}}{25}\right) + \frac{2e^{-5t}}{125} - 0 + (t) \left(\frac{e^0}{25}\right) - \frac{2e^0}{125} \\ &= \frac{e^{-5t}.5t + 2e^{-5t} + 5t - 2}{125} \end{aligned}$$

36. Using Convolution find Inverse Laplace Transforms of: $\frac{1}{(s-2)^4(s+3)}$

[M14/ChemBiot/4M][N14/ChemBiot/7M][N15/CompIT/6M]

[M16/AutoMechCivil/6M]

Solution:



$$\text{Let } F(s) = \frac{1}{(s-2)^4},$$

$$G(s) = \frac{1}{s+3}$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{1}{(s-2)^4}\right\},$$

$$L^{-1}\{G(s)\} = L^{-1}\left\{\frac{1}{s+3}\right\}$$

$$f(t) = e^{2t} \frac{t^3}{3!},$$

$$g(t) = e^{-3t}$$

$$f(u) = \frac{e^{2u}u^3}{6},$$

$$g(t-u) = e^{-3(t-u)}$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(u).g(t-u)du$$

$$\begin{aligned} L^{-1}\left\{\frac{1}{(s-2)^4(s+3)}\right\} &= \int_0^t \frac{e^{2u}u^3}{6} \cdot e^{-3t+3u} du \\ &= \frac{e^{-3t}}{6} \int_0^t u^3 e^{5u} du \\ &= \frac{e^{-3t}}{6} \left[u^3 \left(\frac{e^{5u}}{5} \right) - (3u^2) \left(\frac{e^{5u}}{25} \right) + (6u) \left(\frac{e^{5u}}{125} \right) - 6 \left(\frac{e^{5u}}{625} \right) \right]_0^t \\ &= \frac{e^{-3t}}{6} \left[\frac{t^3 e^{5t}}{5} - \frac{3t^2 e^{5t}}{25} + \frac{6te^{5t}}{125} - \frac{6e^{5t}}{625} + \frac{6}{625} \right] \end{aligned}$$

37. Find Inverse Laplace Transforms of: $\frac{1}{(s+4)^2(s-3)}$ by convolution theorem

[N14/CompIT/6M][N17/IT/4M][M18/Biot/8M]

Solution:

$$\text{Let } F(s) = \frac{1}{(s+4)^2},$$

$$G(s) = \frac{1}{s-3}$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{1}{(s+4)^2}\right\},$$

$$L^{-1}\{G(s)\} = L^{-1}\left\{\frac{1}{s-3}\right\}$$

$$f(t) = e^{-4t}t,$$

$$g(t) = e^{3t}$$

$$f(u) = e^{-4u}u,$$

$$g(t-u) = e^{3(t-u)}$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(u).g(t-u)du$$

$$\begin{aligned} L^{-1}\left\{\frac{1}{(s+4)^2(s-3)}\right\} &= \int_0^t e^{-4u} \cdot u \cdot e^{3t-3u} du \\ &= e^{3t} \int_0^t u e^{-7u} du \\ &= e^{3t} \left[u \left(\frac{e^{-7u}}{-7} \right) - (1) \left(\frac{e^{-7u}}{49} \right) \right]_0^t \\ &= e^{3t} \left[-\frac{te^{-7t}}{7} - \frac{e^{-7t}}{49} - 0 + \frac{e^0}{49} \right] \\ &= e^{3t} \left[\frac{1}{49} - \frac{e^{-7t}}{49} - \frac{te^{-7t}}{7} \right] \end{aligned}$$

38. Find Inverse Laplace Transforms of: $\frac{1}{(s-4)^2(s+3)}$ by convolution theorem

[N18/Comp/6M]

Solution:

$$\text{Let } F(s) = \frac{1}{(s-4)^2},$$

$$G(s) = \frac{1}{s+3}$$



$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{1}{(s-4)^2}\right\}, \quad L^{-1}\{G(s)\} = L^{-1}\left\{\frac{1}{s+3}\right\}$$

$$f(t) = e^{4t}t,$$

$$g(t) = e^{-3t}$$

$$f(u) = e^{4u}u,$$

$$g(t-u) = e^{-3(t-u)}$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(u).g(t-u)du$$

$$L^{-1}\left\{\frac{1}{(s-4)^2(s+3)}\right\} = \int_0^t e^{4u}.u.e^{-3t+3u}du$$

$$= e^{-3t} \int_0^t u e^{7u} du$$

$$= e^{-3t} \left[u \left(\frac{e^{7u}}{7} \right) - (1) \left(\frac{e^{7u}}{49} \right) \right]_0^t$$

$$= e^{-3t} \left[\frac{te^{7t}}{7} - \frac{e^{7t}}{49} - 0 + \frac{e^0}{49} \right]$$

$$= e^{-3t} \left[\frac{1}{49} - \frac{e^{7t}}{49} + \frac{te^{7t}}{7} \right]$$

39. Find $L^{-1}\left\{\frac{1}{(s-2)^3(s+3)}\right\}$ using convolution theorem.

[N19/Comp/6M]

Solution:

$$\text{Let } F(s) = \frac{1}{(s-2)^3},$$

$$G(s) = \frac{1}{s+3}$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{1}{(s-2)^3}\right\},$$

$$L^{-1}\{G(s)\} = L^{-1}\left\{\frac{1}{s+3}\right\}$$

$$f(t) = e^{2t} \frac{t^2}{2!},$$

$$g(t) = e^{-3t}$$

$$f(u) = \frac{e^{2u}u^2}{2},$$

$$g(t-u) = e^{-3(t-u)}$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(u).g(t-u)du$$

$$L^{-1}\left\{\frac{1}{(s-2)^3(s+3)}\right\} = \int_0^t \frac{e^{2u}u^2}{2}.e^{-3t+3u}du$$

$$= \frac{e^{-3t}}{2} \int_0^t u^2 e^{5u} du$$

$$= \frac{e^{-3t}}{2} \left[u^2 \left(\frac{e^{5u}}{5} \right) - (2u) \left(\frac{e^{5u}}{25} \right) + (2) \left(\frac{e^{5u}}{125} \right) \right]_0^t$$

$$= \frac{e^{-3t}}{2} \left[\frac{t^2 e^{5t}}{5} - \frac{2te^{5t}}{25} + \frac{2e^{5t}}{125} - \frac{2}{125} \right]$$

40. Find the inverse LT of $\frac{1}{s(s^2+9)}$ using convolution theorem

[M18/N18/MTRX/4M]

Solution:

$$\text{Let } F(s) = \frac{1}{s},$$

$$G(s) = \frac{1}{s^2+9}$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{1}{s}\right\},$$

$$L^{-1}\{G(s)\} = L^{-1}\left\{\frac{1}{s^2+9}\right\}$$



$$\begin{aligned} f(t) &= 1, & g(t) &= \frac{\sin 3t}{3} \\ f(t-u) &= 1, & g(u) &= \frac{\sin 3u}{3} \end{aligned}$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(t-u).g(u)du$$

$$\begin{aligned} L^{-1}\left\{\frac{1}{s(s^2+9)}\right\} &= \int_0^t 1 \cdot \frac{\sin 3u}{3} du \\ &= \frac{1}{3} \int_0^t \sin 3u du \\ &= \frac{1}{3} \left[-\frac{\cos 3u}{3} \right]_0^t \\ &= \frac{1}{9} [-\cos 3t + \cos 0] \\ &= \frac{1-\cos 3t}{9} \end{aligned}$$

41. Find the inverse LT of $\frac{1}{s(s^2+4)}$ using convolution theorem

[M19/MTRX/4M]

Solution:

$$\begin{aligned} \text{Let } F(s) &= \frac{1}{s}, & G(s) &= \frac{1}{s^2+4} \\ L^{-1}\{F(s)\} &= L^{-1}\left\{\frac{1}{s}\right\}, & L^{-1}\{G(s)\} &= L^{-1}\left\{\frac{1}{s^2+4}\right\} \\ f(t) &= 1, & g(t) &= \frac{\sin 2t}{2} \\ f(t-u) &= 1, & g(u) &= \frac{\sin 2u}{2} \end{aligned}$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(t-u).g(u)du$$

$$\begin{aligned} L^{-1}\left\{\frac{1}{s(s^2+4)}\right\} &= \int_0^t 1 \cdot \frac{\sin 2u}{2} du \\ &= \frac{1}{2} \int_0^t \sin 2u du \\ &= \frac{1}{2} \left[-\frac{\cos 2u}{2} \right]_0^t \\ &= \frac{1}{4} [-\cos 2t + \cos 0] \\ &= \frac{1-\cos 2t}{4} \end{aligned}$$

42. Using Convolution theorem, evaluate $L^{-1}\left\{\frac{1}{(s-1)(s^2+4)}\right\}$

[M14/ElexExtcElectBiomInst/6M]

Solution:

$$\begin{aligned} \text{Let } F(s) &= \frac{1}{s-1}, & G(s) &= \frac{1}{s^2+4} \\ L^{-1}\{F(s)\} &= L^{-1}\left\{\frac{1}{s-1}\right\}, & L^{-1}\{G(s)\} &= L^{-1}\left\{\frac{1}{s^2+4}\right\} \end{aligned}$$



$$f(t) = e^t, \quad g(t) = \frac{\sin 2t}{2}$$

$$f(t-u) = e^{t-u}, \quad g(u) = \frac{\sin 2u}{2}$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(t-u).g(u)du$$

$$L^{-1}\left\{\frac{1}{(s-1)(s^2+4)}\right\} = \int_0^t e^{t-u} \cdot \frac{\sin 2u}{2} du$$

$$= \frac{e^t}{2} \int_0^t e^{-u} \sin 2u du$$

$$= \frac{e^t}{2} \left[\frac{e^{-u}}{(-1)^2+2^2} (-1\sin 2u - 2\cos 2u) \right]_0^t$$

$$= \frac{e^t}{2} \left[\frac{e^{-t}}{5} (-\sin 2t - 2\cos 2t) - \frac{e^0}{5} (-\sin 0 - 2\cos 0) \right]$$

$$= \frac{e^t}{2} \left[-\frac{e^{-t}}{5} (\sin 2t + 2\cos 2t) + \frac{2}{5} \right]$$

$$= \frac{e^t}{5} - \frac{\sin 2t + 2\cos 2t}{10}$$

43. Find Inverse L.T. of: $\frac{1}{(s+3)(s^2+2s+2)}$ by convolution theorem

[N13/Chem/7M]

Solution:

$$\text{Let } F(s) = \frac{1}{s+3}, \quad G(s) = \frac{1}{s^2+2s+2}$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{1}{s+3}\right\}, \quad L^{-1}\{G(s)\} = L^{-1}\left\{\frac{1}{s^2+2s+1+1}\right\}$$

$$f(t) = e^{-3t}, \quad g(t) = L^{-1}\left\{\frac{1}{(s+1)^2+1}\right\}$$

$$f(t) = e^{-3t}, \quad g(t) = e^{-t} \sin t$$

$$f(t-u) = e^{-3(t-u)}, \quad g(u) = e^{-u} \sin u$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(t-u).g(u)du$$

$$L^{-1}\left\{\frac{1}{(s+3)(s^2+2s+2)}\right\} = \int_0^t e^{-3t+3u} \cdot e^{-u} \sin u du$$

$$= e^{-3t} \int_0^t e^{2u} \sin u du$$

$$= e^{-3t} \left[\frac{e^{2u}}{2^2+1} (2\sin u - 1\cos u) \right]_0^t$$

$$= e^{-3t} \left[\frac{e^{2t}}{5} (2\sin t - \cos t) - \frac{e^0}{5} (2\sin 0 - \cos 0) \right]$$

$$= \frac{e^{-3t}}{5} [e^{2t} (2\sin t - \cos t) + 1]$$

44. Find Inverse L.T. of: $\frac{s^2+s}{(s^2+1)(s^2+2s+2)}$ by convolution theorem

[N19/AutoMechCivil/6M]

Solution:



$$\text{Let } F(s) = \frac{s}{s^2+1},$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{s}{s^2+1}\right\},$$

$$f(t) = \cos t,$$

$$f(t) = \cos t,$$

$$f(t-u) = \cos(t-u),$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(t-u).g(u)du$$

$$\begin{aligned} L^{-1}\left\{\frac{s^2+s}{(s^2+1)(s^2+2s+2)}\right\} &= \int_0^t \cos(t-u).e^{-u}\cos u \, du \\ &= \frac{1}{2} \int_0^t e^{-u} 2 \cos(t-u) \cos u \, du \\ &= \frac{1}{2} \int_0^t e^{-u} [\cos(t-u+u) + \cos(t-u-u)] \, du \\ &= \frac{1}{2} \int_0^t e^{-u} \cos t + e^{-u} \cos(t-2u) \, du \\ &= \frac{1}{2} \left[\cos t \cdot \frac{e^{-u}}{-1} + \frac{e^{-u}}{(-1)^2+(-2)^2} (-1 \cdot \cos(t-2u) + (-2) \sin(t-2u)) \right]_0^t \\ &= \frac{1}{2} \left[\frac{e^{-t} \cos t}{-1} + \frac{e^{-t}(-\cos t + 2 \sin t)}{5} - \frac{e^0 \cos t}{-1} - \frac{e^0(-\cos t - 2 \sin t)}{5} \right] \\ &= \frac{1}{10} [e^{-t}(-5 \cos t - \cos t + 2 \sin t) + 5 \cos t + \cos t + 2 \sin t] \\ &= \frac{1}{10} [e^{-t}(2 \sin t - 6 \cos t) + (2 \sin t + 6 \cos t)] \end{aligned}$$

Type V: Use of Differentiation

- Find Inverse Laplace Transforms of: $\log \left[\frac{s+a}{s+b} \right]$

[N19/Chem/4M]

Solution:

$$\begin{aligned} L^{-1}\left\{\log \left(\frac{s+a}{s+b}\right)\right\} &= L^{-1}\{\log(s+a) - \log(s+b)\} \\ &= \frac{-1}{t} L^{-1}\left\{\frac{1}{s+a} - \frac{1}{s+b}\right\} \\ &= \frac{-1}{t} [e^{-at} - e^{-bt}] \\ &= \frac{e^{-bt} - e^{-at}}{t} \end{aligned}$$

- Find Inverse Laplace Transforms of: $\log \left[1 + \frac{a^2}{s^2} \right]$

[N17/IT/4M][M18/N18/Biom/4M]

Solution:

$$\begin{aligned} L^{-1}\left\{\log \left(1 + \frac{a^2}{s^2}\right)\right\} &= L^{-1}\left\{\log \left(\frac{s^2+a^2}{s^2}\right)\right\} \\ &= L^{-1}\{\log(s^2+a^2) - 2 \log s\} \\ &= -\frac{1}{t} L^{-1}\left\{\frac{2s}{s^2+a^2} - \frac{2}{s}\right\} \end{aligned}$$



$$= -\frac{1}{t} [2\cos at - 2]$$

$$= \frac{2}{t} [1 - \cos at]$$

3. Find Inverse L.T. of $\log\left(\frac{s^2+4}{s^2+9}\right)$

[N13/ElexExtcElectBiomInst/4M]

Solution:

$$L^{-1}\left\{\log\left(\frac{s^2+4}{s^2+9}\right)\right\} = L^{-1}\{\log(s^2+4) - \log(s^2+9)\}$$

$$= -\frac{1}{t} L^{-1}\left\{\frac{1}{s^2+4} \times 2s - \frac{1}{s^2+9} \times 2s\right\}$$

$$= -\frac{2}{t} L^{-1}\left\{\frac{s}{s^2+4} - \frac{s}{s^2+9}\right\}$$

$$= -\frac{2}{t} [\cos 2t - \cos 3t]$$

$$= \frac{2(\cos 3t - \cos 2t)}{t}$$

4. Find Inverse Laplace Transforms of: $\log\left[\frac{s^2-1}{s^2}\right]$ Ans. $\frac{2}{t}(1 - \cos ht)$

5. Find Inverse Laplace Transforms of: $\log\left(\frac{s^2+a^2}{\sqrt{s+b}}\right)$

[N13/CompIT/3M][N14/ElexExtcElectBiomInst/4M]

[M17/ElexExtcElectBiomInst/4M]

Solution:

$$L^{-1}\left\{\log\left(\frac{s^2+a^2}{\sqrt{s+b}}\right)\right\} = L^{-1}\{\log(s^2+a^2) - \log\sqrt{s+b}\}$$

$$= L^{-1}\left\{\log(s^2+a^2) - \frac{1}{2}\log(s+b)\right\}$$

$$= -\frac{1}{t} L^{-1}\left\{\frac{1}{s^2+a^2} \times 2s - \frac{1}{2} \cdot \frac{1}{s+b}\right\}$$

$$= -\frac{1}{t} L^{-1}\left\{\frac{2s}{s^2+a^2} - \frac{1}{2(s+b)}\right\}$$

$$= -\frac{1}{t} \left[2\cos at - \frac{e^{-bt}}{2}\right]$$

$$= \frac{1}{t} \left[\frac{e^{-bt}}{2} - 2\cos at\right]$$

6. Find $L^{-1}\{\log(s^4+4)\}$ Ans. $-\frac{4}{t} \cos t \cos ht$

7. Find Inverse Laplace Transforms of: $2 \tanh^{-1} s$

[N14/AutoMechCivil/4M][M18/IT/4M][M18/Elex/3M]

Solution:

$$L^{-1}\{2 \tanh^{-1} s\} = L^{-1}\left\{2 \cdot \frac{1}{2} \log\left(\frac{1+s}{1-s}\right)\right\}$$

$$= L^{-1}\{\log(1+s) - \log(1-s)\}$$

$$= -\frac{1}{t} L^{-1}\left\{\frac{1}{1+s} - \frac{1}{1-s} \times -1\right\}$$

$$= -\frac{1}{t} [e^{-t} - e^t]$$

$$= \frac{e^t - e^{-t}}{t}$$

$$= \frac{2}{t} \sinh t$$

8. Find Inverse Laplace Transforms of: $\frac{1}{s} \log \left(\frac{s+3}{s+2} \right)$

Ans. $\int_0^t \frac{e^{-2u} - e^{-3u}}{u} du$

9. Find $L^{-1} \left\{ \log \left(\frac{s+1}{s-1} \right) \right\}$

[N16/CompIT/4M]

Solution:

$$L^{-1} \left\{ \log \left(\frac{s+1}{s-1} \right) \right\} = L^{-1} \{ \log(s+1) - \log(s-1) \}$$

$$= \frac{-1}{t} L^{-1} \left\{ \frac{1}{s+1} - \frac{1}{s-1} \right\}$$

$$= \frac{-1}{t} [e^{-t} - e^t]$$

$$= \frac{e^t - e^{-t}}{t}$$

$$= \frac{2 \sinh t}{t}$$

10. Find $L^{-1} \left[\log \left(\frac{s^2+1}{s(s+1)} \right) \right]$

[M16/ElexExtcElectBiomInst/4M]

Solution:

$$L^{-1} \left\{ \log \left(\frac{s^2+1}{s(s+1)} \right) \right\} = L^{-1} \{ \log(s^2+1) - \log s - \log(s+1) \}$$

$$= -\frac{1}{t} L^{-1} \left\{ \frac{2s}{s^2+1} - \frac{1}{s} - \frac{1}{s+1} \right\}$$

$$= -\frac{1}{t} [2 \cos t - 1 - e^{-t}]$$

$$= \frac{[1 + e^{-t} - 2 \cos t]}{t}$$

11. Find Inverse Laplace Transforms of: $\log \left[\frac{s-2}{s-3} \right]$

Ans. $\frac{e^{3t} - e^{2t}}{t}$

12. Find Inverse Laplace Transforms of: $\log \left[\frac{s+3}{s+5} \right]$

[N18/IT/4M]

Solution:

$$L^{-1} \left\{ \log \left(\frac{s+3}{s+5} \right) \right\} = L^{-1} \{ \log(s+3) - \log(s+5) \}$$

$$= \frac{-1}{t} L^{-1} \left\{ \frac{1}{s+3} - \frac{1}{s+5} \right\}$$

$$= \frac{-1}{t} [e^{-3t} - e^{-5t}]$$

$$= \frac{e^{-5t} - e^{-3t}}{t}$$

13. Find $L^{-1} \left\{ \log \left[1 + \frac{1}{s^2} \right] \right\}$ Ans. $\frac{2(1-\cos t)}{t}$
14. Find Inverse Laplace Transforms of: $\log \left(\sqrt{\frac{s^2+b^2}{s^2+a^2}} \right)$ Ans. $\frac{\cos bt - \cos at}{t}$
15. Find Inverse Laplace Transforms of: $\log \left(\frac{s^2+a^2}{s^2+b^2} \right)$

[N15/ChemBiot/4M][M18/Biot/6M][M19/Elex/3M]

Solution:

$$\begin{aligned} L^{-1} \left\{ \log \left(\frac{s^2+a^2}{s^2+b^2} \right) \right\} &= L^{-1} \{ \log(s^2 + a^2) - \log(s^2 + b^2) \} \\ &= -\frac{1}{t} L^{-1} \left\{ \frac{1}{s^2+a^2} \times 2s - \frac{1}{s^2+b^2} \times 2s \right\} \\ &= -\frac{2}{t} L^{-1} \left\{ \frac{s}{s^2+a^2} - \frac{s}{s^2+b^2} \right\} \\ &= -\frac{2}{t} [\cos at - \cos bt] \\ &= \frac{2(\cos bt - \cos at)}{t} \end{aligned}$$

16. Find Inverse Laplace Transforms of: $\log \left(\frac{s^2+64}{s^2+16} \right)$

[N17/Elect/4M]

Solution:

$$\begin{aligned} L^{-1} \left\{ \log \left(\frac{s^2+64}{s^2+16} \right) \right\} &= L^{-1} \{ \log(s^2 + 64) - \log(s^2 + 16) \} \\ &= -\frac{1}{t} L^{-1} \left\{ \frac{1}{s^2+64} \times 2s - \frac{1}{s^2+16} \times 2s \right\} \\ &= -\frac{2}{t} L^{-1} \left\{ \frac{s}{s^2+64} - \frac{s}{s^2+16} \right\} \\ &= -\frac{2}{t} [\cos 8t - \cos 4t] \\ &= \frac{2(\cos 4t - \cos 8t)}{t} \end{aligned}$$

17. Find Inverse Laplace Transforms of: $\log \left(\frac{s^2+81}{s^2+36} \right)$

[N17/Biom/4M]

Solution:

$$\begin{aligned} L^{-1} \left\{ \log \left(\frac{s^2+81}{s^2+36} \right) \right\} &= L^{-1} \{ \log(s^2 + 81) - \log(s^2 + 36) \} \\ &= -\frac{1}{t} L^{-1} \left\{ \frac{1}{s^2+81} \times 2s - \frac{1}{s^2+36} \times 2s \right\} \\ &= -\frac{2}{t} L^{-1} \left\{ \frac{s}{s^2+81} - \frac{s}{s^2+36} \right\} \\ &= -\frac{2}{t} [\cos 9t - \cos 6t] \\ &= \frac{2}{t} [\cos 6t - \cos 9t] \end{aligned}$$

18. Find Inverse Laplace Transforms of: $\log\left(\frac{s^2+16}{s^2+81}\right)$

[M19/Inst/4M]

Solution:

$$\begin{aligned} L^{-1}\left\{\log\left(\frac{s^2+16}{s^2+81}\right)\right\} &= L^{-1}\{\log(s^2+16) - \log(s^2+81)\} \\ &= -\frac{1}{t}L^{-1}\left\{\frac{1}{s^2+16} \times 2s - \frac{1}{s^2+81} \times 2s\right\} \\ &= -\frac{2}{t}L^{-1}\left\{\frac{s}{s^2+16} - \frac{s}{s^2+81}\right\} \\ &= -\frac{2}{t}[\cos 4t - \cos 9t] \\ &= \frac{2}{t}[\cos 9t - \cos 4t] \end{aligned}$$

19. Find inverse LT of $\log\left(\frac{1+s^2}{4+s^2}\right)$

[N19/Elex/3M]

Solution:

$$\begin{aligned} L^{-1}\left\{\log\left(\frac{s^2+1}{s^2+4}\right)\right\} &= L^{-1}\{\log(s^2+1) - \log(s^2+4)\} \\ &= -\frac{1}{t}L^{-1}\left\{\frac{1}{s^2+1} \times 2s - \frac{1}{s^2+4} \times 2s\right\} \\ &= -\frac{2}{t}L^{-1}\left\{\frac{s}{s^2+1} - \frac{s}{s^2+4}\right\} \\ &= -\frac{2}{t}[\cos t - \cos 2t] \\ &= \frac{2}{t}[\cos 2t - \cos t] \end{aligned}$$

20. Find inverse LT of $\log\left(\frac{s^2-4}{s^2+121}\right)$

[N18/Inst/4M]

Solution:

$$\begin{aligned} L^{-1}\left\{\log\left(\frac{s^2-4}{s^2+121}\right)\right\} &= L^{-1}\{\log(s^2-4) - \log(s^2+121)\} \\ &= -\frac{1}{t}L^{-1}\left\{\frac{1}{s^2-4} \times 2s - \frac{1}{s^2+121} \times 2s\right\} \\ &= -\frac{2}{t}L^{-1}\left\{\frac{s}{s^2-4} - \frac{s}{s^2+121}\right\} \\ &= -\frac{2}{t}[\cosh 2t - \cos 11t] \\ &= \frac{2}{t}[\cos 11t - \cosh 2t] \end{aligned}$$

21. Find Inverse Laplace Transform of: $\log\left(\frac{s^2+4}{s+4}\right)$

[N18/Extc/4M]

Solution:

$$L^{-1}\left\{\log\left(\frac{s^2+4}{s+4}\right)\right\} = L^{-1}\{\log(s^2+4) - \log(s+4)\}$$



$$\begin{aligned}
 &= -\frac{1}{t} L^{-1} \left\{ \frac{1}{s^2+4} \times 2s - \frac{1}{s+4} \right\} \\
 &= -\frac{1}{t} L^{-1} \left\{ \frac{2s}{s^2+4} - \frac{1}{s+4} \right\} \\
 &= -\frac{1}{t} [2\cos 2t - e^{-4t}] \\
 &= \frac{2}{t} [e^{-4t} - 2\cos 2t]
 \end{aligned}$$

22. Find Inverse Laplace Transforms of: $\log \left[\frac{s^2+a^2}{(s+b)^2} \right]$

[N18/Elect/4M][M19/Elect/4M]

Solution:

$$\begin{aligned}
 L^{-1} \left\{ \log \left[\frac{s^2+a^2}{(s+b)^2} \right] \right\} &= L^{-1} \{ \log(s^2 + a^2) - 2 \log(s + b) \} \\
 &= -\frac{1}{t} L^{-1} \left\{ \frac{2s}{s^2+a^2} - \frac{2}{s+b} \right\} \\
 &= -\frac{1}{t} [2\cos at - 2e^{-bt}] \\
 &= \frac{2}{t} (e^{-bt} - \cos at)
 \end{aligned}$$

23. Find Inverse Laplace Transforms of: $\frac{1}{s} \log \left(\frac{s+2}{s+3} \right)$ Ans. $\int_0^t \frac{e^{-3u} - e^{-2u}}{u} du$

24. Find Inverse Laplace Transforms of: $\frac{1}{s} \log \left(\frac{1+s}{s^2} \right)$ Ans. $\int_0^t \frac{2-2\cos u + \sin u}{u} du$

25. Find Inverse Laplace Transforms of: $\frac{1}{s} \log \left(1 + \frac{1}{s^2} \right)$

[N13/Biot/4M][N16/AutoMechCivil/4M]

Solution:

$$\text{Let } F(s) = \frac{1}{s},$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{1}{s}\right\},$$

$$f(t) = 1,$$

$$f(t-u) = 1,$$

$$G(s) = \log \left(1 + \frac{1}{s^2} \right)$$

$$L^{-1}\{G(s)\} = L^{-1}\left\{\log \left(\frac{s^2+1}{s^2} \right)\right\}$$

$$g(t) = L^{-1}\{\log(s^2 + 1) - 2\log s\}$$

$$g(t) = -\frac{1}{t} L^{-1}\left\{\frac{2s}{s^2+1} - \frac{2}{s}\right\}$$

$$g(t) = -\frac{1}{t} [2\cos t - 2]$$

$$g(t) = \frac{2}{t} [1 - \cos t]$$

$$g(u) = \frac{2}{u} [1 - \cos u]$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(t-u).g(u)du$$

$$\begin{aligned}
 L^{-1}\left\{\frac{1}{s} \log \left(1 + \frac{1}{s^2} \right)\right\} &= \int_0^t 1 \cdot \frac{2}{u} [1 - \cos u] du \\
 &= \int_0^t \frac{2(1-\cos u)}{u} du
 \end{aligned}$$



26. Find Inverse Laplace Transforms of: $\frac{1}{s} \log \left(\frac{s^2+a^2}{s^2+b^2} \right)$

$$\text{Ans. } \int_0^t \frac{2}{u} [\cos bu - \cos au] du$$

27. Using Convolution Theorem, prove that $L^{-1} \left[\frac{1}{s} \log \left(\frac{s+1}{s+2} \right) \right] = \int_0^t \frac{e^{-2u} - e^{-u}}{u} du$

28. Find Inverse Laplace Transforms of: $\frac{1}{s} \log \left(\frac{s+2}{s+1} \right)$

[M18/Elect/4M][M18/Inst/4M]

Solution:

$$\text{Let } F(s) = \frac{1}{s},$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{1}{s}\right\},$$

$$f(t) = 1,$$

$$f(t-u) = 1,$$

$$G(s) = \log \left(\frac{s+2}{s+1} \right)$$

$$L^{-1}\{G(s)\} = L^{-1}\left\{\log \left(\frac{s+2}{s+1} \right)\right\}$$

$$g(t) = L^{-1}\{\log(s+2) - \log(s+1)\}$$

$$g(t) = -\frac{1}{t} L^{-1}\left\{\frac{1}{s+2} - \frac{1}{s+1}\right\}$$

$$g(t) = -\frac{1}{t} [e^{-2t} - e^{-t}]$$

$$g(t) = \frac{e^{-t} - e^{-2t}}{t}$$

$$g(u) = \frac{e^{-u} - e^{-2u}}{u}$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(t-u).g(u)du$$

$$\begin{aligned} L^{-1}\left\{\frac{1}{s} \log \left(1 + \frac{1}{s^2} \right)\right\} &= \int_0^t 1. \frac{e^{-u} - e^{-2u}}{u} du \\ &= \int_0^t \frac{e^{-u} - e^{-2u}}{u} du \end{aligned}$$

29. Find Inverse Laplace Transforms of: $\frac{1}{s} \tanh^{-1} s$

[M14/ElexExtcElectBiomInst/4M]

Solution:

$$\text{Let } F(s) = \frac{1}{s},$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{1}{s}\right\},$$

$$f(t) = 1,$$

$$f(t-u) = 1,$$

$$G(s) = \tanh^{-1} s$$

$$L^{-1}\{G(s)\} = L^{-1}\{\tanh^{-1} s\}$$

$$g(t) = L^{-1}\left\{\frac{1}{2} \log \left(\frac{1+s}{1-s} \right)\right\}$$

$$g(t) = \frac{1}{2} L^{-1}\{\log(1+s) - \log(1-s)\}$$

$$g(t) = -\frac{1}{2t} L^{-1}\left\{\frac{1}{1+s} - \frac{1}{1-s} \times -1\right\}$$

$$g(t) = -\frac{1}{2t} [e^{-t} - e^t]$$

$$g(t) = \frac{\sinh t}{t}$$

$$g(u) = \frac{\sinh u}{u}$$

By convolution theorem,



$$L^{-1}\{F(s).G(s)\} = \int_0^t f(t-u).g(u)du$$

$$L^{-1}\left\{\frac{1}{s} \tanh^{-1} s\right\} = \int_0^t 1. \frac{\sinh u}{u} du = \int_0^t \frac{\sinh u}{u} du$$

30. Show that $L^{-1}\left\{\tan^{-1} \frac{a}{s}\right\} = \frac{\sin at}{t}$

[M16/CompIT/4M]

Solution:

$$\begin{aligned} L^{-1}\left\{\tan^{-1} \frac{a}{s}\right\} &= -\frac{1}{t} L^{-1}\left\{\frac{1}{1+\frac{a^2}{s^2}} \times -\frac{a}{s^2}\right\} \\ &= \frac{a}{t} L^{-1}\left\{\frac{s^2}{s^2+a^2} \times \frac{1}{s^2}\right\} \\ &= \frac{a}{t} L^{-1}\left\{\frac{1}{s^2+a^2}\right\} \\ &= \frac{a}{t} \left[\frac{\sin at}{a}\right] \\ &= \frac{\sin at}{t} \end{aligned}$$

31. Find Inverse Laplace Transforms of: $\tan^{-1} \left[\frac{s+a}{b}\right]$

[N15/AutoMechCivil/3M][N16/ChemBiot/4M][N18/Elex/4M]

[N19/Extc/4M]

Solution:

$$\begin{aligned} L^{-1}\left\{\tan^{-1} \left(\frac{s+a}{b}\right)\right\} &= e^{-at} L^{-1}\left\{\tan^{-1} \left(\frac{s}{b}\right)\right\} \\ &= e^{-at} \cdot -\frac{1}{t} L^{-1}\left\{\frac{1}{1+\left(\frac{s}{b}\right)^2} \times \frac{1}{b}\right\} \\ &= -\frac{e^{-at}}{t} L^{-1}\left\{\frac{b^2}{b^2+s^2} \times \frac{1}{b}\right\} \\ &= -\frac{e^{-at}}{t} L^{-1}\left\{\frac{b}{s^2+b^2}\right\} \\ &= -\frac{e^{-at} \sin bt}{t} \end{aligned}$$

32. Find Inverse Laplace Transforms of: $\tan^{-1} \frac{2}{s^2}$

[N14/ChemBiot/6M][N15/CompIT/6M][M19/IT/6M]

Solution:

$$\begin{aligned} L^{-1}\left\{\tan^{-1} \frac{2}{s^2}\right\} &= -\frac{1}{t} L^{-1}\left\{\frac{1}{1+\left(\frac{2}{s^2}\right)^2} \times 2. \left(-\frac{2}{s^3}\right)\right\} \\ &= \frac{4}{t} L^{-1}\left\{\frac{s^4}{s^4+4} \times \frac{1}{s^3}\right\} \\ &= \frac{4}{t} L^{-1}\left\{\frac{s}{s^4+4}\right\} \\ &= \frac{4}{t} L^{-1}\left\{\frac{s}{s^4+4s^2+4-4s^2}\right\} \\ &= \frac{4}{t} L^{-1}\left\{\frac{s}{(s^2+2)^2-(2s)^2}\right\} \end{aligned}$$



$$\begin{aligned}
 &= \frac{1}{t} L^{-1} \left\{ \frac{4s}{(s^2+2-2s)(s^2+2+2s)} \right\} \\
 &= \frac{1}{t} L^{-1} \left\{ \frac{1}{s^2+2-2s} - \frac{1}{s^2+2+2s} \right\} \\
 &= \frac{1}{t} L^{-1} \left\{ \frac{1}{s^2-2s+1+1} - \frac{1}{s^2+2s+1+1} \right\} \\
 &= \frac{1}{t} L^{-1} \left\{ \frac{1}{(s-1)^2+1} - \frac{1}{(s+1)^2+1} \right\} \\
 &= \frac{1}{t} [e^t \sin t - e^{-t} \sin t] \\
 &= \frac{1}{t} [e^t - e^{-t}] \sin t \\
 &= \frac{2 \sin t \sinh t}{t}
 \end{aligned}$$

33. Using Convolution Theorem, prove that $L^{-1} \left[\frac{1}{s} \tan^{-1} \frac{2}{s} \right] = \int_0^t \frac{1}{u} \sin 2u du$

[M19/Biom/3M]

Solution:

$$\text{Let } F(s) = \frac{1}{s},$$

$$L^{-1}\{F(s)\} = L^{-1}\left\{\frac{1}{s}\right\},$$

$$f(t) = 1,$$

$$f(t-u) = 1,$$

$$G(s) = \tan^{-1} \frac{2}{s}$$

$$L^{-1}\{G(s)\} = L^{-1}\left\{\tan^{-1} \frac{2}{s}\right\}$$

$$g(t) = -\frac{1}{t} L^{-1} \left\{ \frac{1}{1+\frac{2^2}{s^2}} \times -\frac{2}{s^2} \right\}$$

$$g(t) = \frac{2}{t} L^{-1} \left\{ \frac{s^2}{s^2+4} \times \frac{1}{s^2} \right\}$$

$$g(t) = \frac{2}{t} L^{-1} \left\{ \frac{1}{s^2+4} \right\}$$

$$g(t) = \frac{2}{t} \left[\frac{\sin 2t}{2} \right]$$

$$g(u) = \frac{\sin 2u}{u}$$

By convolution theorem,

$$L^{-1}\{F(s).G(s)\} = \int_0^t f(t-u).g(u)du$$

$$L^{-1}\left\{\frac{1}{s} \tan^{-1} \frac{2}{s}\right\} = \int_0^t 1. \frac{\sin 2u}{u} du = \int_0^t \frac{\sin 2u}{u} du$$

34. Find Inverse Laplace Transforms of: $\tan^{-1} \frac{2}{s}$

[N17/Elex/3M]

Solution:

$$L^{-1} \left\{ \tan^{-1} \frac{2}{s} \right\} = -\frac{1}{t} L^{-1} \left\{ \frac{1}{1+\frac{2^2}{s^2}} \times -\frac{2}{s^2} \right\}$$

$$= \frac{2}{t} L^{-1} \left\{ \frac{s^2}{s^2+4} \times \frac{1}{s^2} \right\}$$

$$= \frac{2}{t} L^{-1} \left\{ \frac{1}{s^2+4} \right\}$$

$$= \frac{2}{t} \left[\frac{\sin 2t}{2} \right] = \frac{\sin 2t}{t}$$

35. Find inverse LT of $\tan^{-1}(s + 1)$

[N17/Inst/4M]

Solution:

$$\begin{aligned} L^{-1}\{\tan^{-1}(s + 1)\} &= e^{-t} \cdot L^{-1}\{\tan^{-1} s\} \\ &= e^{-t} \cdot \frac{1}{t} L^{-1}\left\{\frac{1}{s^2+1}\right\} \\ &= -\frac{e^{-t}}{t} [\sin t] \\ &= -\frac{e^{-t} \sin t}{t} \end{aligned}$$

36. Find Inverse Laplace Transforms of: $\cot^{-1} s$

[M17/AutoMechCivil/4M]

Solution:

$$\begin{aligned} L^{-1}\{\cot^{-1} s\} &= \frac{-1}{t} L^{-1}\left\{-\frac{1}{s^2+1}\right\} \\ &= \frac{1}{t} [\sin t] \\ &= \frac{\sin t}{t} \end{aligned}$$

37. Find Inverse Laplace Transforms of: $\cot^{-1}(s + 1)$

[N16/ElexExtcElectBiomInst/4M][M18/N18/MTRX/4M]

Solution:

$$\begin{aligned} L^{-1}\{\cot^{-1}(s + 1)\} &= e^{-t} \cdot L^{-1}\{\cot^{-1} s\} \\ &= e^{-t} \cdot \frac{1}{t} L^{-1}\left\{-\frac{1}{s^2+1}\right\} \\ &= \frac{e^{-t}}{t} [\sin t] \\ &= \frac{e^{-t} \sin t}{t} \end{aligned}$$

38. Find $L^{-1}\left[\cot^{-1}\left(\frac{s+3}{2}\right)\right]$

[N15/ElexExtcElectBiomInst/4M]

Solution:

$$\begin{aligned} L^{-1}\left\{\cot^{-1}\left(\frac{s+3}{2}\right)\right\} &= e^{-3t} L^{-1}\left\{\cot^{-1}\left(\frac{s}{2}\right)\right\} \\ &= e^{-3t} \cdot -\frac{1}{t} L^{-1}\left\{-\frac{1}{1+\left(\frac{s}{2}\right)^2} \times \left(\frac{1}{2}\right)\right\} \\ &= \frac{e^{-3t}}{2t} L^{-1}\left\{\frac{4}{4+s^2}\right\} \\ &= \frac{e^{-3t}}{t} L^{-1}\left\{\frac{2}{s^2+4}\right\} \\ &= e^{-3t} \frac{\sin 2t}{t} \end{aligned}$$

39. Find Inverse Laplace Transforms of: $\cot^{-1}\left(\frac{2}{s^2}\right)$ Ans. $-\frac{2s\sin t \sinh t}{t}$
40. Using Convolution Theorem, prove that $L^{-1}\left[\frac{1}{s}\tan^{-1}\frac{a}{s}\right] = \int_0^t \frac{1}{u} \sin au du$
41. Find Inverse Laplace Transform of $\frac{s^2+5}{(s^2+4s+13)^2}$

[M14/AutoMechCivil/5M]

Solution:

$$\begin{aligned} L^{-1}\left\{\frac{s^2+5}{(s^2+4s+13)^2}\right\} &= L^{-1}\left\{\frac{s^2+5}{(s^2+4s+4+9)^2}\right\} \\ &= L^{-1}\left\{\frac{s^2+5}{((s+2)^2+9)^2}\right\} \\ &= L^{-1}\left\{\frac{(s+2-2)^2+5}{((s+2)^2+9)^2}\right\} \\ &= e^{-2t} L^{-1}\left\{\frac{(s-2)^2+5}{(s^2+9)^2}\right\} \\ &= e^{-2t} L^{-1}\left\{\frac{s^2-4s+9}{(s^2+9)^2}\right\} \\ &= e^{-2t} L^{-1}\left\{\frac{1}{s^2+9} - \frac{4s}{(s^2+9)^2}\right\} \end{aligned}$$

Consider,

$$\begin{aligned} L^{-1}\left\{\frac{1}{s^2+9}\right\} &= \frac{\sin 3t}{3} \\ -\frac{1}{t} L^{-1}\left\{-\frac{1}{(s^2+9)^2} \times 2s\right\} &= \frac{\sin 3t}{3} \\ \frac{2}{t} L^{-1}\left\{\frac{s}{(s^2+9)^2}\right\} &= \frac{\sin 3t}{3} \\ \therefore L^{-1}\left\{\frac{s}{(s^2+9)^2}\right\} &= \frac{t \sin 3t}{6} \end{aligned}$$

Thus,

$$\begin{aligned} L^{-1}\left\{\frac{s^2+5}{(s^2+4s+13)^2}\right\} &= e^{-2t} L^{-1}\left\{\frac{1}{s^2+9} - \frac{4s}{(s^2+9)^2}\right\} \\ &= e^{-2t} \left[L^{-1}\left\{\frac{1}{s^2+9}\right\} - 4 L^{-1}\left\{\frac{s}{(s^2+9)^2}\right\} \right] \\ &= e^{-2t} \left[\frac{\sin 3t}{3} - 4 \frac{t \sin 3t}{6} \right] \\ &= \frac{e^{-2t}}{3} (1 - 2t) \sin 3t \end{aligned}$$